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Introduction to Topology, Second Edition

I. Metric Spaces

1. Open and Closed Sets

1. Let U, V, and W be subsets of some set. Recall that $U \setminus V$ consists of all points in U that do not belong to V.

(a) Prove that $(U \cup V) \setminus W = (U \setminus W) \cup (V \setminus W)$.

(b) Prove that $(U \cap V) \setminus W = (U \setminus W) \cap (V \setminus W)$.

(c) Does $U \setminus (V \setminus W)$ coincide with $(U \setminus V) \setminus W$? Justify your answer by proof or counterexample.

(d) Prove the two set-theoretic identities used in the proof of Theorem 1.9.

2. Show that (1.7) defines a metric on X. Show that every subset of the resulting metric space is both open and closed.

3. (a) Show that if $a, b, c \in \mathbb{R}$ are such that for all $\lambda \in \mathbb{R}$, $a\lambda^2 + b\lambda + c \geq 0$, then $b^2 - 4ac \leq 0$. *Hint:* Find the minimal value of the polynomial in λ.

(b) Show that for any $(x_1, \ldots, x_n), (y_1, \ldots, y_n) \in \mathbb{R}^n$,

$$\sum_{i=1}^{n} x_i y_i \leq \left(\sum_{i=1}^{n} x_i^2 \right)^{1/2} \left(\sum_{i=1}^{n} y_i^2 \right)^{1/2}.$$

Remark: This is a version of the important *Cauchy–Schwarz inequality.* (Perhaps the most common spelling error in mathematics is to replace "Schwarz" by "Schwartz.") For the proof, apply part (a) to $\sum_{i=1}^{n} (x_i - \lambda y_i)^2$.

(c) Using the Cauchy–Schwarz inequality, show that the function d defined by (1.5) satisfies the triangle inequality.

4. Show that the semiopen interval $(0, 1]$ is neither open nor closed in $\mathbb{R}$.

5. Show that (1.6) defines a metric on the space $B(S)$ of bounded real-valued functions on a set S.

6. Let (X, d) be a metric space. Prove that the interior of a subset Y of X coincides with the union of all open subsets of X that are contained in Y. (Thus the interior of Y is the largest open set contained in Y.)

7. Let (X, d) be a metric space. Prove that the closure of a subset Y of X coincides with the intersection of all closed subsets of X that contain Y. (Thus the closure of Y is the smallest closed set containing Y.)

8. Let (X, d) be a metric space. A set of the form $\{y \in X : d(x, y) \leq r\}$ is called a *closed ball*. Show that a closed ball is a closed set. Is the closed ball $\{y \in X : d(x, y) \leq r\}$ always the closure of the open ball $B(x; r)$? What if $X = \mathbb{R}^n$? Prove your answers.

9. Let (X, d) be a metric space. Let Y be a subspace of X and let S be a subset of Y. Show that the closure of S in Y coincides with $\overline{S} \cap Y$, where $\overline{S}$ is the closure of S in X.

10. Let (X, d) be a metric space. A point $x \in X$ is a *limit point* of a subset S of X if every ball $B(x; r)$ contains infinitely many points of S. Show that x is a limit point of S if and only if there is a sequence $\{x_j\}_{j=1}^{\infty}$ in S such that $x_j \to x$ and $x_j \neq x$ for all j. Show that the set of limit points of S is closed.

11. Let (X, d) be a metric space. A point $x \in S$ is an *isolated point* of S if there exists $r > 0$ such that $B(x; r) \cap S = \{x\}$. Show that the closure of a subset S of X is the disjoint union of the limit points of S and the isolated points of S.

12. Let (X, d) be a metric space. Two metrics on X are *equivalent* if they determine the same open subsets. Show that two metrics d, ρ on X are equivalent if and only if the convergent sequences in (X, d) are the same as the convergent sequences in (X, ρ).

13. Let (X, d) be a metric space. Define ρ on $X \times X$ by

$$\rho(x, y) = \min(1, d(x, y)), \qquad x, y \in X.$$

Show that ρ is a metric that is equivalent to d. (Hence every metric is equivalent to a bounded metric.)

14. Let (X, d) be a metric space. The *boundary* ∂E of a set E is defined to be the set of points adherent to both E and the complement of E,

$$\partial E = \overline{E} \cap \overline{(X \setminus E)}.$$

Show that E is open if and only if $E \cap \partial E$ is empty. Show that E is closed if and only if $\partial E \subseteq E$.

2. Completeness

1. A sequence $\{x_k\}_{k=1}^{\infty}$ in a metric space (X, d) is a *fast Cauchy sequence* if

$$\sum_{k=1}^{\infty} d(x_k, x_{k+1}) < \infty.$$

Show that a fast Cauchy sequence is a Cauchy sequence.

2. Prove that every Cauchy sequence has a subsequence that is a fast Cauchy sequence.

3. Prove that the set of isolated points of a countable complete metric space X forms a dense subset of X.

4. Suppose that F is a subset of the first category in a metric space X and E is a subset of F. Prove that E is of the first category in X. Show by an example that E may not be of the first category in the metric space F.

5. Prove that any countable union of sets of the first category in X is again of the first category in X.

6. Let S be a nonempty set, let (X, d) be a metric space, and let $\mathcal{F}$ be the set of functions from S to X. For $f, g \in \mathcal{F}$, define

$$\rho(f, g) = \sup_{s \in S} \min(1, d(f(s), g(s))).$$

Show that ρ is a metric in $\mathcal{F}$. Show that a sequence $\{f_n\}$ converges to f in the metric space $(\mathcal{F}, \rho)$ if and only if $\{f_n\}$ converges uniformly to f on X. Show that $(\mathcal{F}, \rho)$ is complete if and only if (X, d) is complete.

7. Let (X, d) be a metric space and let S be the set of Cauchy sequences in S. Define a relation "$\sim$" in X by declaring "$\{s_k\} \sim \{t_k\}$" to mean that $d(s_k, t_k) \to 0$ as $k \to \infty$.

(a) Show that the relation "$\sim$" is an equivalence relation.

(b) Let $\widetilde{X}$ denote the set of equivalence classes of S and let $\overline{s}$ denote the equivalence class of $s = \{s_k\}_{k=1}^{\infty}$. Show that the function

$$\rho(\overline{s}, \overline{t}) = \lim_{k\to\infty} d(s_k, t_k), \qquad \overline{s}, \overline{t} \in \widetilde{X}$$

defines a metric on $\widetilde{X}$.

(c) Show that $(\widetilde{X}, \rho)$ is complete.

(d) For $x \in X$, define $\overline{x}$ to be the equivalence class of the constant sequence $\{x, x, \ldots\}$. Show that the function $x \mapsto \overline{x}$ is an isometry of X onto a dense subset of $\widetilde{X}$. (By an isometry, we mean that $d(x, y) = \rho(\overline{x}, \overline{y})$, $x, y \in X$.)

Note: If a complete metric space Y contains X as a dense subspace, we say that Y is a *completion* of X. The space $\widetilde{X}$ of Exercise 7 can be regarded as a completion of X by identifying each $x \in X$ with the constant sequence $\{x, x, \ldots\}$. The next part of the exercise shows that the completion of X is unique, up to isometry.

(e) Show that when Y is a completion of X, then the inclusion map $X \to Y$ extends to an isometry of $\widetilde{X}$ onto Y.

8. The *diameter* of a nonempty subset E of a metric space (X, d) is defined to be

$$\operatorname{diam}(E) = \sup\{d(x, y) : x, y \in E\}.$$

Show that if $\{E_k\}_{k=1}^{\infty}$ is a decreasing sequence of closed nonempty subsets of a complete metric space whose diameters tend to zero, then $\bigcap_{k=1}^{\infty} E_k$ consists of precisely one point. How much of the conclusion remains true if X is not complete? Can this property be used to characterize complete metric spaces? Justify your answer.

3. The Real Line

1. Use the Least Upper Bound Axiom, together with the usual manipulations of algebraic identities and inequalities, to prove the following:

(a) The set $\mathbb{Z}$ of integers is not bounded above.

(b) For each $\varepsilon > 0$, there exists a rational number $r \in (0, \varepsilon)$.

(c) If $a, b \in \mathbb{R}$ satisfy $a < b$, then there exists a rational number $s \in (a, b)$.

(d) The set $\mathbb{R}_0$ of rational numbers is dense in $\mathbb{R}$.

2. Prove that the set of irrational numbers is dense in $\mathbb{R}$.

3. Regard the rational numbers $\mathbb{R}_0$ as a subspace of $\mathbb{R}$. Does the metric space $\mathbb{R}_0$ have any isolated points? Why does this not contradict Exercise 2.3?

4. Prove that every open subset of $\mathbb{R}$ is a union of disjoint open intervals (finite, semi-infinite, or infinite).

5. Prove that the set of irrational numbers cannot be expressed as the union of a sequence of closed subsets of $\mathbb{R}$.

6. Prove that the set of rational numbers cannot be expressed as the intersection of a sequence of open subsets of $\mathbb{R}$.

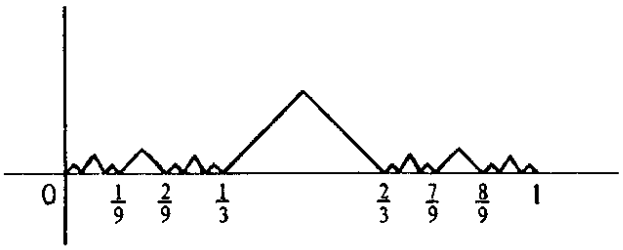
7. Let E_0 be the closed unit interval $[0, 1]$. Let E_1 be the closed subset of E_0 obtained by removing the open, middle third of E_0, so that $E_1 = [0, 1/3] \cup [2/3, 1]$. Let E_2 be the closed subset of E_1 obtained by removing the open middle thirds of each of the two intervals in E_1, so that E_2 consists of four closed intervals, each of length $1/3^2$. Proceeding in this manner, we construct E_n to consist of 2^n closed intervals, and we obtain E_{n+1} by removing the open middle third from each interval of E_n. The *Cantor set* is defined to be the intersection of the E_n's. Prove the following:

(a) The Cantor set is a closed subset of $\mathbb{R}$ with empty interior.

(b) The Cantor set has no isolated points.

(c) The Cantor set is uncountable. *Hint:* Use Exercise 2.3.

8. Define explicitly a continuous function f on the unit interval $[0, 1]$ which is zero precisely on the Cantor set and which has a graph as suggested by the figure.



Prove that f satisfies the following condition:

$$|f(s) - f(t)| \leq |s - t|, \qquad 0 \leq s, t \leq 1.$$

Remark: This type of condition is called a *Lipschitz condition* on f. Lipschitz conditions will arise again in Section 8.

9. Determine the interior, the closure, the limit points, and the isolated points of each of the following subsets of $\mathbb{R}$:

(a) the interval $[0, 1)$,

(b) the set of rational numbers,

(c) $\{m + n\pi : m \text{ and } n \text{ positive integers}\}$,

(d) $\left\{\dfrac{1}{m} + \dfrac{1}{n} : m \text{ and } n \text{ positive integers}\right\}$.

10. Let f be a real-valued function on $\mathbb{R}$. Show that there exist $M > 0$ and a nonempty open subset U of $\mathbb{R}$ such that for any $s \in U$, there is a sequence $\{s_n\}$ satisfying $s_n \to s$ and $|f(s_n)| \leq M$, $n \geq 1$.

4. Products of Metric Spaces

1. Prove that the functions on $X \times X$ given by (4.1), (4.2), and (4.3) are metrics.

2. For each of the metrics (4.1), (4.2), and (4.3) on the plane $\mathbb{R}^2$, sketch the open ball $B(0;1)$. What inclusion relations hold between these balls?

3. Prove that each of the metrics (4.1), (4.2), and (4.3) has property (4.4).

4. Determine the interior, the closure, the limit points, and the isolated points of each of the following subsets of $\mathbb{R}^2$:

(a) $\{(x, y) : 0 < x^2 + y^2 < 1\}$,

(b) $\{(x, 0) : x \in \mathbb{R}\}$,

(c) $\{(m, n) : m \text{ and } n \text{ positive integers}\}$,

(d) $\left\{\left(\frac{1}{m}, \frac{1}{n}\right) : m \text{ and } n \text{ positive integers}\right\}$.

5. Prove that if $\widetilde{X}_j$ is the completion of X_j, $1 \leq j \leq n$, then $\widetilde{X}_1 \times \cdots \times \widetilde{X}_n$ is the completion of $X_1 \times \cdots \times X_n$.

6. Let (X_k, d_k), $1 \leq k \leq n$, be metric spaces. Let d be a metric on $X = X_1 \times \cdots \times X_n$ such that the open subsets of (X, d) are precisely the unions of sets of the form $U_1 \times \cdots \times U_n$, where U_k is an open subset of X_k, $1 \leq k \leq n$. Show that d satisfies (4.4). *Note:* This establishes the converse to Theorem 4.1.

7. Let (X_k, d_k), $1 \le k < \infty$, be metric spaces and let $X = \prod_{k=1}^{\infty} X_k$ be their Cartesian product, that is, let X be the set of sequences $(x_1, x_2, \ldots)$, where $x_j \in X_j$ for $1 \le j < \infty$.

(a) Show that

$$d(x, y) = \sum_{j=1}^{\infty} \frac{1}{2^j} \min(1, d_j(x_j, y_j))$$

is a metric on X.

(b) Show that a sequence $\{x^{(k)}\}_{k=1}^{\infty}$ converges in X if and only if $\{x_j^{(k)}\}_{k=1}^{\infty}$ converges in X_j for each $j \ge 1$.

(c) Show that the open sets in X are precisely the unions of sets of the form $U_1 \times \cdots \times U_m \times X_{m+1} \times X_{m+2} \times \cdots$, where $m \ge 1$ and U_j is open in X_j, $1 \le j \le m$.

5. Compactness

1. Give an example of a totally bounded metric space which is not compact.

2. Give an example of a complete metric space which is not compact.

3. Show directly that a compact metric space is totally bounded.

4. Let $\{U_\alpha\}_{\alpha \in A}$ be a finite open cover of a compact metric space X.

(a) Show that there exists $\varepsilon > 0$ such that for each $x \in X$, the open ball $B(x; \varepsilon)$ is contained in one of the U_α's.

Remark: Such an ε is called a *Lebesgue number* of the cover.

(b) Show that if at least one of the U_α's is a proper subset of X, then there is a largest Lebesgue number for the cover.

5. Prove that the product of a finite number of compact metric spaces is compact.

6. A metric space is a *Polish space* if it is separable and if there is an equivalent metric for which it is complete. Prove that any closed subspace of a Polish space is Polish.

Remark: An important school of Polish mathematicians made deep contributions to the study of point-set topology during the period between the world wars. The terminology "Polish" recognizes the work of the Polish school on this important class of spaces. For more on Polish spaces, see Exercises 6.14 and 6.15.

7. Let $B(S)$ be the metric space of bounded real-valued functions on the set S, with the metric of uniform convergence on S (defined in Section 1). Show that $B(S)$ is separable if and only if S is finite.

Hint: For any subset T of S, define $\chi_T \in B(S)$ by

$$\chi_T(s) = \begin{cases} 1, & s \in T, \\ 0, & s \notin T. \end{cases}$$

Then the balls $B(\chi_T, 1/2)$ are disjoint.

8. Let (X, d) be a bounded metric space and let $\mathcal{F}$ be the family of nonempty closed subsets of X. Show that

$$\rho(E, F) = \max\left(\sup_{x\in E} d(x, F), \sup_{y\in F} d(y, E)\right)$$

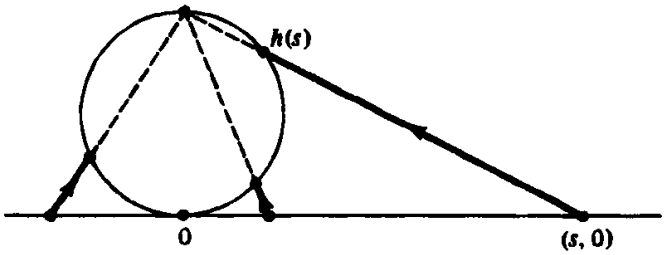
defines a metric on $\mathcal{F}$. Show that $\mathcal{F}$ is compact whenever X is compact.

6. Continuous Functions

1. Show that if (X, d) is a metric space, then d is a continuous function from $X \times X$ to $\mathbb{R}$. Furthermore, show that for each fixed $x_0 \in X$, the function $x \mapsto d(x, x_0)$ is a uniformly continuous function from X to $\mathbb{R}$.

2. Prove that two metrics d and ρ for X are equivalent if and only if the identity map $(X, d) \to (X, \rho)$ is bicontinuous (that is, it and its inverse are continuous).

3. Let E be the subspace of $\mathbb{R}^2$ obtained from the circle centered at $(0, 1/2)$ of radius $1/2$ by deleting the point $(0, 1)$. Define a function h from $\mathbb{R}$ to E so that $h(s)$ is the point at which the line segment from $(s, 0)$ to $(0, 1)$ meets E.



(a) Show that h is a bicontinuous function from $\mathbb{R}$ to E.

(b) Show that

$$\rho(s, t) = |h(s) - h(t)|, \qquad s, t \in \mathbb{R},$$

defines a metric ρ on $\mathbb{R}$ that is equivalent to the usual metric on $\mathbb{R}$.

(c) Show that $(\mathbb{R}, \rho)$ is totally bounded but not complete.

(d) Identify the completion of $(\mathbb{R}, \rho)$.

4. Let f be a continuous function from the metric space X to the metric space Y and let E be a subset of X. Show that the restriction of f to E is a continuous function from E to Y.

5. Let X_0, X_1, and X_2 be metric spaces, let f be a function from X_0 to X_1, and let g be a function from X_1 to X_2. Show that if f is continuous at $x_0 \in X_0$ and if g is continuous at $f(x_0)$, then the composition $g \circ f$ is continuous at x_0.

6. Let f be a continuous function from the closed unit interval $[0, 1]$ to itself. Show that there exists $t \in [0, 1]$ such that $f(t) = t$.

Remark: The point t is called a *fixed point* of f.

7. Show that if f is a continuous function from a compact metric space X to a metric space Y, then the image of f is a compact subset of Y. In particular, show that f is bounded, that is, that the range of f is a bounded subset of Y.

8. Prove that a continuous real-valued function on a compact metric space assumes its maximum value and its minimum value.

9. Prove that a metric space X is compact if and only if every continuous real-valued function on X is bounded.

10. Let (X, d) be a metric space and let S be a nonempty set. Let $B(S, X)$ denote the family of bounded functions from S to X.

(a) Show that

$$\rho(f, g) = \sup_{s \in S} d(f(s), g(s))$$

defines a metric ρ on $B(S, X)$. The metric ρ is called the *metric of uniform convergence* on S.

(b) Show that $B(S, X)$ is complete if and only if X is complete.

(c) Show that if S is itself a metric space, then the space $BC(S, X)$ of bounded continuous functions from S to X is a closed subspace of $B(S, X)$.

11. Show that there exists a continuous real-valued function h on $[0,1]$ such that

$$\limsup_{t\to 0+} \frac{|h(x+t)-h(x)|}{t} = \infty$$

whenever $0 \le x < 1$.

Hint: Consider the space $C([0,2])$ of continuous real-valued functions on the interval $[0,2]$, with the metric of uniform convergence. Let E_m be the set of $f \in C([0,2])$ for which there exists $x \in [0,1]$ satisfying $|f(x+t)-f(x)|/|t| \le m$ for $t > 0$, $x + t \le 2$. Show that E_m is a closed nowhere-dense subset of $C([0,2])$.

12. Let X and Y be metric spaces such that X is complete. Show that if $\{f_\alpha\}_{\alpha \in A}$ is a family of continuous functions from X to Y such that $\{f_\alpha(x) : \alpha \in A\}$ is a bounded subset of Y for each $x \in X$, then there exists a nonempty open subset U of X such that $\{f_\alpha(x) : \alpha \in A, x \in U\}$ is a bounded subset of Y.

13. Let (X, d) and (Y, ρ) be metric spaces such that X is complete. Let $\{g_k\}_{k=1}^{\infty}$ be a sequence of continuous functions from X to Y such that $\{g_k(x)\}$ converges for each $x \in X$. Show that for each $\varepsilon > 0$, there exist $N \geq 1$ and a nonempty open subset U of X such that $\rho(g_n(x), g_m(x)) < \varepsilon$ for all $x \in U$ and all $n, m \geq N$.

14. (a) Prove that if X is a metric space and U is an open subset of X, then there is a positive continuous function f on U such that $f(x) \to +\infty$ as x tends to any point of ∂U.

Hint: Set $f(x) = 1/\inf\{d(x,y) : y \in X \setminus U\}$.

(b) Prove that an open subset of a Polish space is Polish.

Remark: Polish spaces were defined in Exercise 5.6. For the proof, consider a metric on U of the form $\rho(x,y) = d(x,y) + |f(x) - f(y)|$, where f is as in part (a).

(c) Prove that if $\{U_j\}_{j=1}^{\infty}$ is a sequence of open subsets of a Polish space, then $\bigcap U_j$ is a Polish space.

Hint: Extend your proof of (b) by considering

$$\rho(x,y) = d(x,y) + \sum_{j=1}^{\infty} \frac{1}{2^j} \frac{|f_j(x) - f_j(y)|}{1 + |f_j(x) - f_j(y)|},$$

where f_j "blows up" at ∂U_j, as in (a).

(d) Prove that the subset of $\mathbb{R}$ consisting of the irrational numbers is a Polish space.

Remark: It can be shown that a subspace of a Polish space is Polish if and only if it is the intersection of a sequence of open subsets.

15. Let N be the space of positive integers and let $N^\infty = N \times N \times \cdots$ be the countable product space, endowed with the metric introduced in Exercise 4.7.

(a) Show that N^∞ is a Polish space.

(b) Show that if X is any Polish space, there is a continuous function f from N^∞ into X such that f maps open sets onto open sets.

Hint: Construct by induction open balls $B_{n_1 n_2 \cdots n_m}$ in X such that

(i) $B_{n_1 \cdots n_m} \subseteq B_{n_1 \cdots n_{m-1}}$.

(ii) $\bigcup_{n_m=1}^{\infty} B_{n_1 \cdots n_m} = B_{n_1 \cdots n_{m-1}}$.

(iii) the radius of $B_{n_1 \cdots n_m}$ is less than $1/m$.

Remark: In Chapter II, Section 13, the notion of a quotient topological space will be defined and studied. Part (b) of this exercise shows that every Polish space is a quotient space of N^∞.

7. Normed Linear Spaces

1. Sketch the unit ball of $\mathbb{R}^2$ when it is endowed with each of the following norms: $\|\cdot\|_1$, $\|\cdot\|_2$, $\|\cdot\|_4$, and $\|\cdot\|_\infty$. Sketch the ball of radius 1 and center $(2, 1)$ in the metric determined by each of these norms.

2. Show the following for $x \in \mathbb{R}^n$:

(a) $\|x\|_\infty \leq \|x\|_p$, $1 \leq p < \infty$,

(b) $\|x\|_p \leq n^{1/p}\|x\|_\infty$, $1 \leq p < \infty$,

(c) $\lim_{p\to\infty} \|x\|_p = \|x\|_\infty$.

3. Two norms $\|\cdot\|$ and $|||\cdot|||$ on a vector space $\mathfrak{X}$ are *equivalent* if there exists $c > 0$ such that

$$\frac{1}{c}\,|||x||| \le \|x\| \le c\,|||x|||, \qquad x \in \mathfrak{X}.$$

(a) Show that the norms $\|\cdot\|$ and $|||\cdot|||$ are equivalent if and only if the identity map of $\mathfrak{X}$ is bicontinuous between the $\|\cdot\|$-topology of $\mathfrak{X}$ and the $|||\cdot|||$-topology of $\mathfrak{X}$.

(b) Show that the norms $\|\cdot\|$ and $|||\cdot|||$ are equivalent if and only if $\sup\{|||x||| : \|x\| = 1\} < \infty$ and $\inf\{|||x||| : \|x\| = 1\} > 0$.

(c) Prove that the notion of being equivalent norms is an equivalence relation on the set of norms on $\mathfrak{X}$.

4. Show that any two norms on $\mathbb{R}^n$ are equivalent.

Hint: For $x = (x_1, \ldots, x_n)$, show that $\|x\| \leq c\|x\|_1$, where $\|\cdot\|$ is the given norm and $\|x\|_1 = \sum |x_j|$. Then observe that the unit sphere of $\mathbb{R}^n$ in the $\|\cdot\|_1$-norm is $\|\cdot\|$-compact and hence $\inf\{\|x\| : \|x\|_1 = 1\} > 0$.

5. Prove that any linear functional on a finite dimensional normed space is continuous. Use this result to prove that any linear operator from a finite dimensional normed space to another normed space is continuous.

6. A linear transformation T from a normed space $(\mathfrak{X}, \|\cdot\|)$ to a normed space $(\mathfrak{Y}, |||\cdot|||)$ is an *isometric isomorphism* if T is one-to-one and onto and if

$$|||Tx||| = \|x\|, \qquad x \in \mathfrak{X}.$$

(a) Show that the composition of isometric isomorphisms is an isometric isomorphism.

(b) Show that the inverse of an isometric isomorphism is an isometric isomorphism.

(c) Prove that a one-to-one linear transformation T from $\mathfrak{X}$ to $\mathfrak{Y}$ is an isometric isomorphism if and only if the image under T of the unit ball of $\mathfrak{X}$ coincides with the unit ball of $\mathfrak{Y}$.

7. Fix $1 < p < \infty$ and let $q = p/(p-1)$ be the *conjugate index* of p, so that $(1/p) + (1/q) = 1$.

(a) Show that if $s, t \geq 0$, then

$$st \leq \frac{s^p}{p} + \frac{t^q}{q}.$$

(b) Show that if $x, y \in \mathbb{R}^n$, then

$$\left| \sum x_j y_j \right| \leq \|x\|_p \|y\|_q.$$

Hint: First assume $\|x\|_p = 1 = \|y\|_q$.

(c) Show that $\| \cdot \|_p$ is a norm on $\mathbb{R}^n$.

Hint: Write

$$\sum |x_j + y_j|^p = \sum |x_j||x_j + y_j|^{p-1} + \sum |y_j||x_j + y_j|^{p-1}$$

and apply the result of (b) to each summand.

8. Show the following:

(a) ℓ^p is a Banach space, $1 \leq p \leq \infty$.

(b) If $1 \leq p < r < \infty$, then ℓ^p is a proper dense subset of ℓ^r.

(c) ℓ^1 is not dense in ℓ^∞. (Can you identify the closure of ℓ^1 in ℓ^∞?)

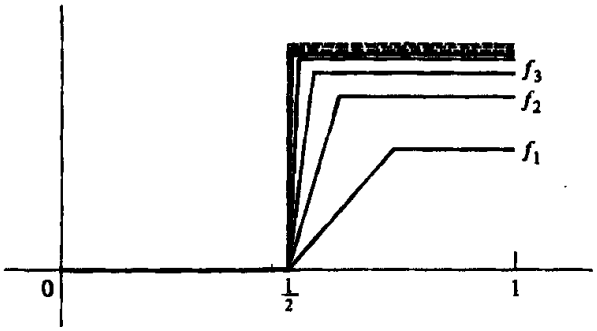
9. For each $s = (s_1, s_2, \ldots) \in \ell^\infty$, show that

$$L_s(x) = \sum_{j=1}^{\infty} s_j x_j, \qquad x \in \ell^1,$$

defines a continuous linear functional on ℓ^1. Show that the correspondence $s \mapsto L_s$ is an isometric isomorphism of ℓ^∞ and the conjugate space $(\ell^1)^*$ of ℓ^1. (This type of theorem is called a *representation theorem*, in that it represents the conjugate space of a Banach space as some more concrete Banach space.)

10. Show that $\|f\|_1 = \int_0^1 |f(s)|\, ds$ defines a norm on $f \in C[0,1]$.

11. Show that the space of continuous real-valued functions on the unit interval $[0, 1]$ is not complete in the norm $\|\cdot\|_1$ defined earlier.



Hint: Consider a sequence of functions whose graphs are as indicated in the figure.

12. (a) Prove that if f and g are continuous real-valued functions on $[0,1]$, then

$$\int_0^1 f(s)g(s)\,ds \le \left(\int_0^1 f(s)^2\,ds\right)^{1/2}\left(\int_0^1 g(s)^2\,ds\right)^{1/2}.$$

Remark: This is an integral version of the Cauchy–Schwarz inequality. For the proof, refer back to Exercise 1.3 and consider the integral

$$\int_0^1 [f(s)-\lambda g(s)]^2\,ds,$$

which is nonnegative for all values of the real parameter λ.

(b) Using (a), show that the formula

$$\|f\|_2 = \left(\int_0^1 f(s)^2\,ds\right)^{1/2}$$

defines a norm on the space of continuous real-valued functions on $[0,1]$.

(c) Prove that if f and g are continuous complex-valued functions on $[0,1]$, then

$$\int_0^1 f(s)\overline{g(s)}\,ds \le \left(\int_0^1 |f(s)|^2\,ds\right)^{1/2}\left(\int_0^1 |g(s)|^2\,ds\right)^{1/2}.$$

Remark: This is an integral form of the complex version of the Cauchy–Schwarz inequality.

(d) Show that the formula

$$\|f\|_2 = \left(\int_0^1 |f(s)|^2\,ds\right)^{1/2}$$

defines a norm on the space of continuous complex-valued functions on $[0,1]$.

13. Show that there is a one-to-one correspondence between norms on a vector space $\mathfrak{X}$ and metrics d on $\mathfrak{X}$ that are translation-invariant and homogeneous in the sense that

$$d(x + w, y + w) = d(x, y), \qquad x, y, w \in \mathfrak{X},$$

and

$$d(cx, cy) = |c|d(x, y), \qquad x, y \in \mathfrak{X},\ c \text{ scalar}.$$

14. Let $\mathfrak{Y}$ be the vector space of all real-valued functions on a nonempty set S. Show that

$$d(f,g) = \sup_{s\in S} \frac{|f(s) - g(s)|}{1 + |f(s) - g(s)|}$$

is a metric on $\mathfrak{Y}$ that is translation-invariant. Does the metric arise from a norm?

15. Prove that a sequence in $\mathfrak{B}(\mathfrak{X}, \mathfrak{Y})$ converges to $T \in \mathfrak{B}(\mathfrak{X}, \mathfrak{Y})$ in the uniform operator topology if and only if it converges to T uniformly on each bounded subset of $\mathfrak{X}$.

16. Let $\mathfrak{X}$, $\mathfrak{Y}$, and $\mathfrak{Z}$ be normed spaces, let $S \in \mathfrak{B}(\mathfrak{X}, \mathfrak{Y})$, and let $T \in \mathfrak{B}(\mathfrak{Y}, \mathfrak{Z})$. Show that the composed operator TS from $\mathfrak{X}$ to $\mathfrak{Z}$ is bounded and satisfies

$$\|TS\| \leq \|T\|\|S\|.$$

17. Let $\mathfrak{X}$ and $\mathfrak{Y}$ be Banach spaces and let $\mathfrak{X} \oplus \mathfrak{Y}$ be their vector space direct sum. Show that

$$\|(x, y)\| = \max(\|x\|, \|y\|)$$

defines a norm on $\mathfrak{X} \oplus \mathfrak{Y}$ that makes $\mathfrak{X} \oplus \mathfrak{Y}$ into a Banach space.

18. Let $\mathfrak{X}$ and $\mathfrak{Y}$ be Banach spaces and let $\{T_n\}$ be a sequence in $\mathfrak{B}(\mathfrak{X}, \mathfrak{Y})$ such that $\{T_n x\}$ converges for all $x \in \mathfrak{X}$. Show that there exists $T \in \mathfrak{B}(\mathfrak{X}, \mathfrak{Y})$ such that $\lim T_n x = Tx$ for all $x \in \mathfrak{X}$.

19. Consider the linear functional L on the real Banach space $C[0,1]$, endowed with the sup norm, defined by

$$L(f) = \sum_{j=1}^{m} a_j f(t_j),$$

where the t_j's are distinct points of $[0,1]$ and the a_j's are real numbers. Show that L is continuous, with norm

$$\|L\| = \sum_{j=1}^{m} |a_j|.$$

20. For each integer $n \geq 0$, let $\{t_j^{(n)}\}_{j=1}^{m_n}$ be points in $[0,1]$ and let $\{a_j^{(n)}\}_{j=1}^{m_n}$ be positive real numbers such that

$$\int_0^1 g(t)\,dt = \sum_{j=1}^{m_n} a_j^{(n)} g(t_j^{(n)})$$

for all polynomials g of degree at most n. Show that

$$\int_0^1 f(t)\,dt = \lim_{n\to\infty} \sum_{j=1}^{m_n} a_j^{(n)} f(t_j^{(n)})$$

for all $f \in C[0,1]$. *Hint:* Use Theorem 7.6.

21. Let $h \in C[0,1]$, sup norm. Define a linear functional L on $C[0,1]$ by

$$L(f) = \int_0^1 f(t)h(t)\,dt, \qquad f \in C[0,1].$$

What is $\|L\|$? Prove your answer.

22. Let $k(s,t)$ be a continuous function on $[0,1] \times [0,1]$. Let T be the linear operator on $C[0,1]$ endowed with the sup norm, defined by

$$(Tf)(s) = \int_0^1 k(s,t)f(t)\,dt, \qquad 0 \le s \le 1.$$

What is $\|T\|$? Prove your answer.

Remark: The function $k(s,t)$ is called a *kernel function*; it is said to be the kernel of the integral operator T.

8. The Contraction Principle

1. Show that if Φ is a function from a nonempty compact metric space X to itself such that

$$d(\Phi(x), \Phi(y)) < d(x, y), \qquad x \neq y,$$

then Φ has a unique fixed point.

2. Find a compact metric space X and a function $\Phi : X \to X$ such that

$$d(\Phi(x), \Phi(y)) \leq d(x, y), \qquad x, y \in X, \tag{8.13}$$

while Φ has no fixed point. Find also X and Φ satisfying (8.13) such that Φ has more than one fixed point.

3. Let X be a complete metric space and let Φ be a function from X to X such that Φ^m is a contraction for some $m \geq 1$. Show that Φ has a unique fixed point.

4. Let $\mathfrak{X}$ be a Banach space, let $m \geq 1$, and let T be a continuous linear operator on $\mathfrak{X}$ such that $\|T^m\| < 1$. Fix $u \in \mathfrak{X}$ and define

$$\Phi(v) = u + T(v), \qquad v \in \mathfrak{X}.$$

(a) Show that Φ^m is a contraction.

(b) Show that the equation

$$v = u + T(v)$$

has a unique solution $v \in \mathfrak{X}$.

5. Let $\mathfrak{X}$ be a Banach space and let T be a continuous linear operator on $\mathfrak{X}$. Let $u \in \mathfrak{X}$, let $\lambda \in \mathbb{C}$, and define $\Phi(v) = (u/\lambda) + T(v/\lambda)$, $v \in \mathfrak{X}$.

(a) By applying the Contraction Mapping Theorem to Φ, show that the equation

$$\lambda v = u + T(v) \tag{8.14}$$

has a unique solution $v \in \mathfrak{X}$, provided $|\lambda| > \|T\|$.

(b) Let $v_0 = 0$ and $v_m = \Phi(v_{m-1})$, $m \geq 1$. Show that

$$v_m = \sum_{k=0}^{m+1} \frac{T^k(u)}{\lambda^{k+1}}.$$

(c) Show that if $|\lambda| > \|T\|$, then the series

$$\sum_{k=0}^{\infty} T^k/\lambda^{k+1} \tag{8.15}$$

converges in norm to $(\lambda I - T)^{-1}$. How can this be used to solve (8.14)?

(d) Show that if

$$|\lambda| > \limsup_{n\to\infty} \|T^n\|^{1/n},$$

then the series (8.15) converges in norm to $(\lambda I - T)^{-1}$.

6. Consider the Volterra integral equation

$$v(s) = u(s) + \int_a^s K(s,t)v(t)\,dt, \qquad a \le s \le b, \tag{8.16}$$

where K is continuous on $[a,b] \times [a,b]$. Show that (8.16) has a unique solution $v \in C[a,b]$ for each fixed $u \in C[a,b]$. *Hint:* Show that the integral operator T appearing in (8.16) satisfies

$$\|T^m\| \le M^m(b-a)^m/m!, \qquad m \ge 0,$$

where M is a bound for the kernel function K. Then apply Exercise 5 or Exercise 4.

7. Prove in detail the uniqueness assertion of the Cauchy–Picard Theorem.

8. Show that the initial-value problem

$$\begin{cases} \dfrac{du}{dt} = u^{1/2}, & t \geq 0, \\ u(0) = 0 \end{cases}$$

has more than one solution. (*Hint:* Look for solutions of the form $u(t) = t^{\alpha}$.) Reconcile this fact with the Cauchy–Picard Theorem.

9. Formulate and prove a theorem to the effect that the solution of the initial-value problem (8.6) depends continuously on the initial data ξ.

10. Let $F : U \times [a, b] \to \mathbb{R}^n$ be continuous and suppose that F satisfies locally a Lipschitz condition in the first variable, i.e., for each $\xi \in U$, there is a ball $B(\xi; r) \subseteq U$ on which F satisfies the Lipschitz condition (8.8), with c depending possibly on ξ and r. Show that either there is a continuously differentiable solution of (8.6) on $[a, b]$ or else there exists β, $a < \beta \leq b$, and a continuously differentiable solution u of (8.6) on $[a, \beta)$ such that $u(t) \to (\partial U) \cup \{\infty\}$ as t increases to β.

11. Let X, S, Φ_s, and $x_s^* = x^*(s)$ be as in Theorem 8.5. Fix $x_0 \in X$ and define $x_m(s) = \Phi_s^m(x_0)$ to be the mth iterate of x_0 under Φ_s. Show that if S is compact, then x_m converges uniformly on S to x^*.

9. The Fréchet Derivative

1. Define $G : \mathbb{R}^2 \to \mathbb{R}^1$ by $G(0,0) = 0$ and $G(x,y) = xy/(x^2+y^2)^{1/2}$ if $(x,y) \neq (0,0)$. Show that G is continuous and the partial derivatives of G with respect to x and y exist everywhere, but G is not Fréchet differentiable at $(0,0)$.

2. Prove that if $G = (G_1, \ldots, G_m)$ maps $\mathbb{R}^n$ to $\mathbb{R}^m$ and if the partial derivatives $\partial G_j / \partial x_k$, $1 \leq j \leq m$, $1 \leq k \leq n$, exist everywhere and are continuous, then G is continuously differentiable in the sense of Fréchet.

3. Prove that if $F : \mathfrak{X} \to \mathfrak{Y}$ is differentiable at $x_0 \in \mathfrak{X}$ and if $G : \mathfrak{Y} \to \mathfrak{Z}$ is differentiable at $y_0 = F(x_0) \in \mathfrak{Y}$, then $G \circ F$ is differentiable at x_0 and

$$(G \circ F)'(x_0) = G'(F(x_0))F'(x_0).$$

4. Prove or disprove that

$$\|A^{-1}\| = 1/\|A\|.$$

5. Prove that the Fréchet derivative of the operator function $A \mapsto A^{-1}$, defined on the invertible operators from $\mathfrak{X}$ to $\mathfrak{Y}$, is continuous.

6. Fix an interval $[a, b]$ and let F be a continuously differentiable function on $\mathbb{R} \times [a, b]$. Define a nonlinear integral operator Ψ on the Banach space $C[a, b]$ of continuous real-valued functions on $[a, b]$ by

$$(\Psi u)(t) = \int_a^t F(u(s), s)\, ds, \qquad a \le t \le b.$$

Prove that Ψ is differentiable in the sense of Fréchet and that the derivative $\Psi'(u_0)$ of Ψ at $u_0 \in C[a, b]$ is the Volterra integral operator T defined by

$$(Tv)(t) = \int_a^t F_1(u_0(s), s)v(s)\, ds, \qquad a \le t \le b,$$

where F_1 is the partial derivative of F with respect to the first variable.

7. Let F be a continuously differentiable real-valued function on $\mathbb{R} \times [a, b]$ and consider the initial-value problem

$$\begin{cases} \dfrac{du}{dt} = F(u, t), \quad a \le t \le b, \\ u(a) = \xi, \end{cases} \tag{9.12}$$

treated in Section 8. The existence of a unique solution $u_\xi \in C[a, b]$ is guaranteed by the Cauchy–Picard Theorem (Theorem 8.4). Define a function Φ from $\mathbb{R} \times C[a, b]$ to $C[a, b]$ by

$$\Phi(\xi, u)(t) = u(t) - \xi - \int_a^t F(u(s), s)\, ds, \qquad a \le t \le b.$$

(a) Identify the partial derivatives Φ_1 and Φ_2 of Φ.

(b) Show that Φ satisfies the hypotheses of the Implicit Function Theorem. *Hint:* Use Exercise 6 and Exercise 8.6.

(c) Show that the solution u_ξ of (9.12) depends continuously on the initial value ξ.

8. Let U be an open subset of a Banach space $\mathfrak{X}$ and let F be a continuously differentiable function from U to a Banach space $\mathfrak{Y}$ such that $F'(x)$ is invertible for all $x \in U$. Show that F is an open map, that is, that the image under F of any open subset of U is an open subset of $\mathfrak{Y}$.

9. A function F from $\mathbb{R}^l$ to $\mathbb{R}^m$ is a C^k-function if each of the component functions of F has continuous partial derivatives of all orders $\leq k$. Suppose that F is a C^k-function ($k \geq 1$) from $\mathbb{R}^n \times \mathbb{R}^m$ to $\mathbb{R}^m$ such that $F(0,0) = 0$ and the Jacobian matrix

$$\left(\frac{\partial F_i}{\partial y_j}(0,0)\right)_{i,j=1}^{m}$$

is invertible. Prove that there is a C^k-function f, defined in a neighborhood U of 0 in $\mathbb{R}^n$, such that $F(x, f(x)) = 0$, $x \in U$.

10. Let $k \geq 2$ and let f be a real-valued C^k-function, defined on the interval $(-\delta, \delta)$ in $\mathbb{R}$, such that $f(0) = 0 = f'(0)$. Let $M = \{(t, f(t)) : |t| < \delta\}$ be the graph of f in $\mathbb{R}^2$. Let $h(x, y)$ denote the distance from a point $(x, y) \in \mathbb{R}^2$ to M, defined by

$$h(x, y)^2 = \inf\{(t - x)^2 + (f(t) - y)^2 : |t| < \delta\}.$$

(a) Prove that there is a neighborhood U of 0 such that each point (x, y) of U has a unique nearest point $(t(x, y), f(t(x, y)))$ in M.

(b) Prove that $t(x, y)$ is a C^{k-1}-function on U.

(c) Prove that h^2 is a C^k-function on U and h is a C^k-function in $U \setminus M$.

(d) Show that if $f(t) = |t|^{3/2}$, then points $(0, y)$, $y > 0$, do not have a unique nearest point in M and, moreover, h is not a C^1-function near the semiaxis $\{(0, y) : y > 0\}$.

(e) Extend parts (a), (b), and (c) to functions f from $\mathbb{R}^n$ to $\mathbb{R}^m$.

Hint: The distance-squared function $|(x, y) - (t, f(t))|^2$ is minimized where its t-derivative is zero. Thus $t(x, y)$ is obtained by solving implicitly the equation

$$F(x, y, t) = t - x + [f(t) - y]f'(t) = 0.$$

The Implicit Function Theorem yields a C^{k-1} solution of this equation (Exercise 9). To show that h^2 is a C^k-function requires an additional computation:

$$\frac{\partial}{\partial x}h^2 = 2(x - t(x, y)), \qquad \frac{\partial}{\partial y}h^2 = 2(y - f(t(x, y))).$$

II. Topological Spaces

1. Topological Spaces

1. Show that the intersection of a family of topologies for X is a topology for X.

2. Let X be a set and let $\mathcal{T}$ be the family of subsets U of X such that $X \setminus U$ is finite, together with the empty set $\varnothing$. Show that $\mathcal{T}$ is a topology. ($\mathcal{T}$ is called the *cofinite topology* of X.)

3. Let $\mathcal{T}$ be the cofinite topology on the set $\mathbb{Z}$ of integers. Show that the sequence $\{1, 2, 3, \ldots\}$ converges in $(\mathbb{Z}, \mathcal{T})$ to each point of $\mathbb{Z}$. Describe the convergent sequences in $(\mathbb{Z}, \mathcal{T})$.

4. Let S be a subset of a set X. Describe the closure of S when (i) X has the discrete topology, (ii) X has the indiscrete topology, and (iii) X has the cofinite topology.

5. Let S be a subset of a topological space X in which sets consisting of one point are closed. A point $x \in X$ is a *limit point* of S if every neighborhood of x contains a point of S other than x itself. A point $s \in S$ is an *isolated point* of S if there is a neighborhood U of s such that $U \cap S = \{s\}$. Show that the set of limit points of S is closed. Show that $\overline{S}$ is the disjoint union of the set of limit points of S and the isolated points of S.

6. (a) Show that if X is a metrizable topological space and if p and q are distinct points of X, then there are open sets U and V such that $p \in U$, $q \in V$, and $U \cap V = \varnothing$.

(b) Let X be an infinite set and let $\mathcal{T}$ be the cofinite topology on X (Exercise 2). Prove that the property described in part (a) does not hold for the open sets in X and hence that X with the cofinite topology is not metrizable.

7. Show that if S is a subset of a topological space X, then $\text{int}(S)$ is the union of all open sets contained in S.

8. Show that if S is a subset of a topological space X, then $\overline{S}$ is the intersection of all closed sets containing S.

9. Show that if S is a subset of a topological space X, then

(a) $\overline{X \setminus S} = X \setminus \text{int}(S)$,

(b) $\text{int}(X \setminus S) = X \setminus \overline{S}$.

10. If U is open, then prove that $U^{-'-'-} = U^{-}$, where the bar means *closure* and the prime means *complement*.

11. Prove the following result, originally noted by Kuratowski: If S is a subset of a topological space X, then there are at most 14 subsets of X that can be obtained from S by successively taking either complements or closures. Find a subset S of $\mathbb{R}$ such that exactly fourteen subsets of $\mathbb{R}$ can be obtained from S in this manner.

Hint: For the proof, first observe that $S'' = S$ and $S^{--} = S^{-}$, so that we need consider only sets of the form $S'^{-\prime-\cdots}$ and $S^{-\prime-\prime\cdots}$. Now apply Exercise 10. For the example, use a subset of $\mathbb{R}$ with different pieces separated from each other: an open interval, a half-open interval, a sequence converging to a point, etc.

12. Let X be a set and let $\mathcal{T}$ be the family of subsets U of X such that $X \setminus U$ is at most countable, together with the empty set $\varnothing$.

(a) Prove that $\mathcal{T}$ is a topology for X.

(b) Describe the convergent sequences in X with respect to this topology.

(c) Prove that if X is uncountable, then there is a subset S of X whose closure contains points that are not limits of convergent sequences in S. *Hint:* Take S to be any proper uncountable subset of X.

2. Subspaces

1. Let X be a topological space, let S be a subspace of X, and let E be a subset of S. Show that the relative topology that E inherits from S coincides with the relative topology that E inherits from X.

2. Prove that if A and S are subsets of a topological space X, then the closure of $A \cap S$ in S in the relative topology for S is a subset of the intersection $\overline{A} \cap S$, where $\overline{A}$ is the closure of A in X. Give an example where the relative closure of $A \cap S$ is a proper subset of $\overline{A} \cap S$.

3. Let S be a subset of a topological space X. Show that a sequence $\{x_i\}$ in S converges to $x_0 \in S$ in the relative topology if and only if, considered as a sequence in X, the sequence $\{x_i\}$ converges to x_0.

4. Prove that a set S in X is open if and only if every relatively open subset of S is open in X. Is this statement true if "open" is replaced by "closed"?

5. Let X be a topological space and let S be a subset of X. Show that if A is a relatively open subset of S, then $A \cap T$ is a relatively open subset of $S \cap T$ for any subset T of X.

6. Let X be a topological space, let S and T be subsets of X, and let A be a subset of $S \cap T$ that is relatively open in S and in T. Is A relatively open in $S \cup T$? Justify your answer.

3. Continuous Functions

1. Let f be a function from a metric space X to a metric space Y and let $x \in X$. Show that the definition of continuity of f at x given in this section (for each open neighborhood V of $f(x)$, there exists an open neighborhood U of x such that $f(U) \subseteq V$) coincides with the definition given in Section I.6 (whenever $\{x_n\}$ is a sequence in X such that $x_n \to x$, then $f(x_n) \to f(x)$).

2. Show that a function $f : X \to Y$ is continuous if and only if $f^{-1}(E)$ is a closed subset of X for every closed subset E of Y.

3. Prove the following statements about continuous functions and discrete and indiscrete topological spaces.

(a) If X is discrete, then every function f from X to a topological space Y is continuous.

(b) If X is not discrete, then there is a topological space Y and a function $f : X \to Y$ that is not continuous.

Hint: Let Y be the set X with the discrete topology.

(c) If Y is an indiscrete topological space, then every function f from a topological space X to Y is continuous.

(d) If Y is not indiscrete, then there is a topological space X and a function $f : X \to Y$ that is not continuous.

4. Prove that all open intervals in $\mathbb{R}$ (finite, semi-infinite, or infinite) are homeomorphic.

5. Prove that all semiopen intervals in $\mathbb{R}$ (finite or semi-infinite) are homeomorphic.

6. Show that the unit ball $\{(x, y) \in \mathbb{R}^2 : x^2 + y^2 < 1\}$ in $\mathbb{R}^2$ is homeomorphic to the open square $\{(x, y) : 0 < x < 1, 0 < y < 1\}$.

7. Show that the punctured plane $\mathbb{R}^2 \setminus \{(0,0)\}$ is homeomorphic to the exterior of the closed unit ball $\mathbb{R}^2 \setminus \{(x,y) : x^2 + y^2 \leq 1\}$.

8. Show that $\mathbb{R}^2 \setminus \{(x, y) : x \text{ and } y \text{ integers}\}$ is homeomorphic to the space $\mathbb{R}^2 \setminus \{(x, y) :$ there are integers n and m such that $(x - n)^2 + (y - m)^2 < 1/10\}$.

9. Show that the open right half plane $\{(x, y) \in \mathbb{R}^2 : x > 0\}$ is homeomorphic to the unit ball $\{(x, y) \in \mathbb{R}^2 : x^2 + y^2 < 1\}$.

10. Develop analogues for $\mathbb{R}^n$, $n > 2$, of Exercises 6 through 9.

11. Prove that metrizability is a topological property.

12. Prove that if f and g are continuous functions from a topological space X to $\mathbb{R}$, then $f + g$ and fg are also continuous. Furthermore, if $g \neq 0$ on X, then f/g is continuous.

13. Prove that if $f : X \to Y$ is continuous and if S is a subspace of X, then the restriction $f|_S : S \to Y$ is continuous.

14. Let X be a topological space and let $BC(X)$ denote the set of bounded continuous real-valued functions on X, with the metric

$$d(f, g) = \sup\{|f(x) - g(x)| : x \in X\}.$$

Show that $BC(X)$ is a complete metric space.

4. Base for a Topology

1. Let X be a topological space with the cofinite topology (Exercise 1.2). Show that X is separable. When is X second-countable?

2. Let X be a topological space with the discrete topology. Find a base $\mathcal{A}$ of open sets for X such that $\mathcal{A}$ is included in any other base of open sets for X.

3. Let X and Y be topological spaces and let $\mathcal{A}$ be a base of open sets for Y. Show that a function $f : X \to Y$ is continuous if and only if $f^{-1}(U)$ is an open subset of X for every $U \in \mathcal{A}$.

4. A topological space X satisfies the *first axiom of countability*, or is *first-countable*, if for each $x \in X$, there exists a sequence of open neighborhoods $\{U_n\}_{n=1}^{\infty}$ of x such that each neighborhood of x includes one of the U_n's. Prove the following assertions:

(a) Any metric space is first-countable.

(b) Any second-countable space is first-countable.

(c) In a first-countable space, any point adherent to a set S is a limit of a sequence in S.

5. Let X be a set and let $\mathcal{S}$ be a family of subsets of X.

(a) Show that there exists a unique smallest topology $\mathcal{T}$ on X such that $\mathcal{S} \subseteq \mathcal{T}$. The topology $\mathcal{T}$ is called the *topology generated* by $\mathcal{S}$, and $\mathcal{S}$ is called a *subbase* for the topology $\mathcal{T}$.

(b) Let $\mathcal{B}$ be the family of subsets of X consisting of X, $\varnothing$, and all finite intersections of sets in $\mathcal{S}$. Show that $\mathcal{B}$ is a base of open sets for the topology generated by $\mathcal{S}$.

(c) Let X have the topology generated by $\mathcal{S}$, let Y be a topological space, and let $f : Y \to X$ be a function. Show that f is continuous if and only if $f^{-1}(S)$ is an open subset of Y for every set $S \in \mathcal{S}$.

6. Let $\mathcal{B}$ be the family of subsets of $\mathbb{R}$ of the form $[a, b)$, where $-\infty < a < b < \infty$.

(a) Show that $\mathcal{B}$ is a base of open sets for a topology $\mathcal{T}$ of $\mathbb{R}$. The topology determined by $\mathcal{B}$ is the *half-open interval topology*.

(b) Show that every open subset of $\mathbb{R}$ (in the metric topology) is $\mathcal{T}$-open.

(c) Show that each interval $[a, b)$ is $\mathcal{T}$-closed.

(d) Show that a point $t \in \mathbb{R}$ lies in the $\mathcal{T}$-closure of a subset S of $\mathbb{R}$ if and only if there is a sequence $\{t_n\}_{n=1}^{\infty}$ in S such that $t_n \geq t$ and $|t_n - t| \to 0$.

(e) Show that a function f from $(\mathbb{R}, \mathcal{T})$ to $\mathbb{R}$ is continuous if and only if f is continuous from the right at each $t \in \mathbb{R}$, that is, if and only if

$$\lim_{\varepsilon > 0,\, \varepsilon \to 0} f(t + \varepsilon) = f(t), \qquad t \in \mathbb{R}.$$

7. Prove that a subspace of a second-countable space is second-countable.

8. Regard $\mathbb{R}$ as a subset of $\mathbb{R}^2$ in the usual way. Let $\mathcal{T}$ be the family of all subsets U of $\mathbb{R}^2$ such that $U \setminus \mathbb{R}$ is an open subset of $\mathbb{R}^2$ (in the usual topology). Establish the following assertions:

(a) $\mathcal{T}$ is a topology for $\mathbb{R}^2$.

(b) $(\mathbb{R}^2, \mathcal{T})$ is separable.

(c) The subspace $\mathbb{R}$ of $(\mathbb{R}^2, \mathcal{T})$ is neither separable nor second-countable.

(d) $(\mathbb{R}^2, \mathcal{T})$ is not second-countable.

5. Separation Axioms

1. Show that a sequence in a Hausdorff space cannot converge to more than one point.

2. Let X be a topological space and let X_0 be the topological space that is the set X with the cofinite topology. Show that the identity map of X to X_0 is continuous if and only if X is a T_1-space.

3. A property of a topological space is *hereditary* if, whenever a topological space X has that property, then every subspace of X has the property. Show that the properties of being a T_1-space, Hausdorff space, and regular space are hereditary.

4. Prove that a topological space X is normal if and only if the conclusion of Urysohn's Lemma is valid for X. Prove that this occurs if and only if the conclusion of the Tietze Extension Theorem is valid for X.

5. Let X be the subspace of $\mathbb{R}^2$ consisting of the closed upper half-plane minus the origin:

$$X = \{(x, y) : y \geq 0, (x, y) \neq (0, 0)\}.$$

Show that $E = \{(x, 0) : x < 0\}$ and $F = \{(x, 0) : x > 0\}$ are disjoint closed subsets of X. Find a continuous function $f : X \to [0, 1]$ such that $f = 0$ on E and $f = 1$ on F.

6. Let X be a normal topological space, let E be a closed subset of X, and let $f : E \to \mathbb{R}$ be continuous. Show that f can be extended to a continuous function from X to $\mathbb{R}$. *Hint:* Let $g : \mathbb{R} \to (-1, 1)$ be a homeomorphism, extend $g \circ f$ to a map from X to $[-1, 1]$, and decide what to do on the set where the extension assumes the values ± 1.

7. Let X have the cofinite topology. When is X a T_1-space? When is X a Hausdorff space?

8. Let $\mathcal{T}$ be the half-open interval topology for $\mathbb{R}$, defined in Exercise 4.6. Show that $(\mathbb{R}, \mathcal{T})$ is a T_4-space.

9. Let $\mathcal{B}$ be the collection of subsets of $\mathbb{R}$ of the form (a, b), together with those of the form $(a, b) \cap \mathbb{R}_0$, where $-\infty < a < b < \infty$ and $\mathbb{R}_0$ is the set of rational numbers. Prove the following:

(a) $\mathcal{B}$ is a base of open sets for a topology $\mathcal{T}$ for $\mathbb{R}$.

(b) $(\mathbb{R}, \mathcal{T})$ is a Hausdorff space.

(c) $\mathbb{R} \setminus \mathbb{R}_0$ is $\mathcal{T}$-closed.

(d) If $f : (\mathbb{R}, \mathcal{T}) \to \mathbb{R}$ is a continuous function such that $f = 0$ on $\mathbb{R} \setminus \mathbb{R}_0$, then $f = 0$ everywhere.

(e) $(\mathbb{R}, \mathcal{T})$ is not regular.

Remark: Exercise 9 provides an example of a Hausdorff space that is not regular. For an example of a regular Hausdorff space that is not normal, see Exercises 10.2 and 10.5.

6. Compactness

1. Prove in detail that compactness is a topological property.

2. Prove that a topological space X is compact if and only if it has the following property: if $\{E_\alpha\}_{\alpha\in A}$ is any family of closed subsets of X such that any finite intersection of the E_α's is nonempty, then $\bigcap_{\alpha\in A} E_\alpha$ is nonempty.

3. Show that any space with the cofinite topology is compact.

4. Show that a discrete topological space is compact if and only if it is finite.

5. Show that a continuous real-valued function on a compact space attains its maximum value and its minimum value. In particular, show that a continuous real-valued function on a compact space is bounded.

6. Prove that if X is a compact Hausdorff space and if $x, y \in X$ satisfy $x \neq y$, then there is a continuous real-valued function f on X such that $f(x) \neq f(y)$. (In other words, the continuous real-valued functions on X "separate the points" of X.)

7. Let X be a compact Hausdorff space and let $\{U_\alpha\}_{\alpha\in A}$ be an open cover of X. Show that there exist a finite number of continuous real-valued functions $h_1, \ldots, h_m$ on X with the following properties:

(a) $0 \le h_j \le 1$, $1 \le j \le m$,

(b) $\sum_j h_j = 1$,

(c) For each $1 \le j \le m$, there is an index α_j such that the closure of the set $\{x : h_j(x) > 0\}$ is contained in U_{α_j}.

Remark: The h_j's are called a *partition of unity* subordinate to the cover $\{U_\alpha\}$. They partition the function 1 into a number of bump functions, each of which "lives" inside one of the U_α's. For the proof, use the normality to construct, for each $x \in X$, a bump function g_x such that $g_x \ge 0$, $g_x(x) > 0$, and such that the open set $V_x = \{g_x > 0\}$ satisfies $\overline{V}_x \subseteq U_\alpha$ for some $\alpha = \alpha(x)$. Choose $x_1, \ldots, x_m$ such that the V_{x_j}'s cover X, and set $h_j = g_{x_j} / \sum g_{x_i}$.

8. A family F of real-valued functions on a topological space X is *equicontinuous* if for each $x \in X$ and $\varepsilon > 0$, there exists a neighborhood U of x such that $|f(y) - f(x)| < \varepsilon$ for all $y \in U$ and all $f \in F$. Let $\{f_k\}_{k=1}^{\infty}$ be a bounded sequence of real-valued functions on a compact space X such that the $\{f_k\}$ are equicontinuous. Show that there is a uniformly convergent subsequence of $\{f_k\}$. *Hint:* For each $n \geq 1$, find points $x_{n1}, \ldots, x_{nm_n}$ and open neighborhoods W_{nj} of the x_{nj} such that $X = W_{n1} \cup \cdots \cup W_{nm_n}$ and $|f_k(x_{nj}) - f_k(y)| < 1/n$ whenever $y \in W_{nj}$, $1 \leq j \leq m_n$, and $k \geq 1$. Use a diagonalization procedure to find a subsequence $\{f_{k_i}\}$ such that $\{f_{k_i}(x_{nj})\}_{i=1}^{\infty}$ converges for each fixed n and j. Then show that $\{f_{k_i}\}$ converges uniformly on X.

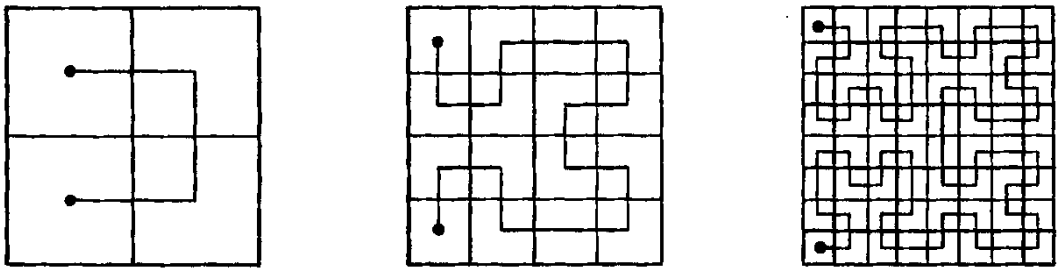
9. Let X be compact and let $C(X)$ denote the set of continuous real-valued functions on X. Show that

$$d(f, g) = \sup\{|f(x) - g(x)| : x \in X\}$$

defines a metric on $C(X)$, so that $C(X)$ becomes a complete metric space.

10. Let X be compact. Prove the Arzelà–Ascoli Theorem, that a subset F of $C(X)$ is a compact subset of $C(X)$ if and only if F is closed, bounded, and equicontinuous. *Hint:* For the "if" half of the proof, use Exercise 8 and Theorem I.5.1.

11. Define functions f_1, f_2, f_3 from $[0,1]$ to the square $[0,1] \times [0,1]$ as suggested by the three diagrams, so that as t runs from 0 to 1 the images $f_j(t)$ run at uniform speed along the polygonal curves.



Approximations to a space-filling curve

Continue in this fashion to obtain $f_j : [0,1] \to [0,1] \times [0,1]$ for all $j \geq 1$.

(a) Prove that the sequence $\{f_j\}$ converges uniformly on $[0,1]$ to some function f.

(b) Prove that the image of f is the square $[0,1] \times [0,1]$. (Such a space-filling curve is called a *Peano curve*; this particular procedure for constructing a space-filling curve is due to Hilbert.)

(c) The kth curve f_k can be traced by a sequence of pencil moves of equal length in one of the four directions up (U), down (D), left (L), or right (R). Find an algorithm for expressing f_k in terms of these simple moves (and in terms of the expression for f_{k-1}).

7. Locally Compact Spaces

1. Prove that a Hausdorff topological space X is locally compact if and only if, for each $p \in X$ and each open neighborhood U of p, there is an open neighborhood V of p such that $V \subseteq U$ and $\overline{V}$ is compact.

2. Prove in detail that local compactness is a topological property.

3. Prove that a locally compact Hausdorff space is regular.

4. Prove that if every point of X has an open neighborhood U such that $\overline{U}$ is a compact Hausdorff space, then X is Hausdorff.

5. Let X be a topological space and define the family $\mathcal{B}$ of subsets of $Y = X \cup \{\infty\}$ as in the proof of Theorem 7.1. Which conclusions of Theorem 7.1 are valid without any further hypotheses on X? Which are valid whenever X is a Hausdorff space?

6. Let X be locally compact Hausdorff space that is not compact. Prove that X is dense in the one-point compactification $Y = X \cup \{\infty\}$ of X. (Thus the one-point compactification is indeed a compactification.)

7. Prove that the one-point compactification of $\mathbb{R}^n$ is homeomorphic to the n-sphere

$$S^n = \{(y_1, \ldots, y_{n+1}) \in \mathbb{R}^{n+1} : y_1^2 + \cdots + y_{n+1}^2 = 1\}.$$

Hint: It suffices to show that $S^n \setminus (0, \ldots, 0, 1)$ is homeomorphic to $\mathbb{R}^n$. See Exercise I.6.3.

8. Prove that the one-point compactification of any open ball in $\mathbb{R}^n$ is homeomorphic to the n-sphere S^n. *Hint:* See Exercise 7.

8. Connectedness

1. Prove in detail that connectedness is a topological property.

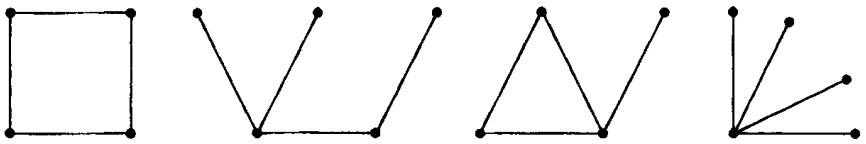
2. Prove that every connected subset of $\mathbb{R}$ is an interval.

3. Consider the two tangent open discs $\{x^2 + y^2 < 1\}$ and $\{(x-2)^2 + y^2 < 1\}$ in $\mathbb{R}^2$. Is the union of the discs a connected subset of $\mathbb{R}^2$? Is the union of their closures a connected subset of $\mathbb{R}^2$? Is the union of one disc and the closure of the other a connected subset of $\mathbb{R}^2$?

4. A point p of a topological space X is a *cut point* if $X \setminus \{p\}$ is disconnected. Show that the property of having a cut point is a topological property.

5. Show that no two of the intervals $[0, 1]$, $(0, 1)$, and $[0, 1)$ are homeomorphic. *Hint:* Use Exercise 4.

6. Each of the topological spaces in the figure below is a union of four closed intervals. Are any of the spaces homeomorphic? Prove your answer. *Hint:* Extend the technique used in Exercise 5.



7. Let f be a continuous one-to-one function from the unit interval $[0, 1]$ into $\mathbb{R}^n$ ($n \geq 2$). Prove that the image of f has no interior. *Hint:* First note that f is a homeomorphism onto its range. Then prove and use the fact that a punctured ball in $\mathbb{R}^n$ is connected when $n \geq 2$.

8. A topological space is *totally disconnected* if the connected components are all singletons. Prove that any countable metric space is totally disconnected.

9. Prove that each connected component of a topological space is closed.

10. Show by counterexample that a connected component of a topological space is not necessarily open. *Hint:* See Exercise 8.

11. A topological space X is *locally connected* if, for each point $p \in X$ and each open set U containing p, there is a connected open set V with $p \in V$ and $V \subseteq U$. Show that each connected component of a locally connected topological space is open.

12. Let $\mathcal{T}$ be the half-open interval topology defined for $\mathbb{R}$ in Exercise 4.6. What are the connected components of $(\mathbb{R}, \mathcal{T})$?

13. What are the connected components of $(\mathbb{R}, \mathcal{T})$ where $\mathcal{T}$ is the topology introduced in Exercise 5.9, that is Hausdorff but not regular?

14. Let X be a locally compact Hausdorff space that is not compact and let $Y = X \cup \{\infty\}$ be its one-point compactification. If X is connected, is Y connected? If Y is connected, is X connected? Justify your answers.

9. Path Connectedness

1. Prove that any subinterval of $\mathbb{R}$ (closed, open, or semiopen) is path-connected.

2. Prove that path connectedness is a topological property.

3. Prove that if X is path-connected and $f : X \to Y$ is a map, then $f(X)$ is path-connected.

4. A space X is *locally path-connected* if, for each open subset V of X and each $x \in V$, there is a neighborhood U of x such that x can be joined to any point of U by a path in V. Prove that the path components of a locally path-connected space coincide with the connected components.

5. Suppose X is locally path-connected and locally connected (as defined in Exercise 8.11).

(a) Show that each path component of X is open and closed.

(b) Show that for each $x \in X$ and each neighborhood V of x, there is an open path-connected neighborhood U of x such that $U \subseteq V$.

6. An open subset of $\mathbb{R}^n$ is connected if and only if it is path-connected. *Hint:* Use Exercise 4.

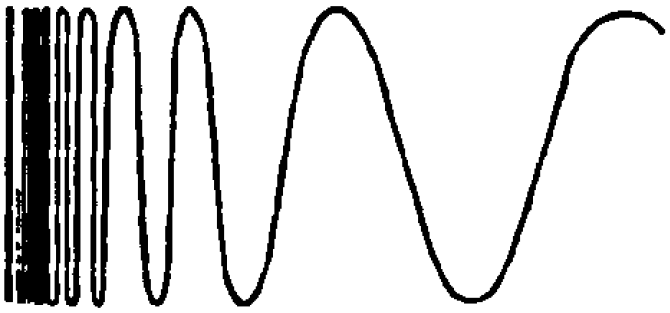
7. Let X be the compact subset of $\mathbb{R}^2$ consisting of the vertical interval

$$E = \{(0, y) : -1 \leq y \leq 1\}$$

together with the portion of the graph of $\sin(1/x)$ given by

$$F = \{(x, \sin(1/x)) : 0 < x \leq 1\}.$$

(See the figure below.) Show that X is connected but not path-connected. What are the path components of X?



Connected but not path connected

8. Show that the path components of X are not necessarily closed subsets of X, nor are they necessarily open. *Hint:* Use the example in Exercise 7.

10. Finite Product Spaces

1. Show that if E_j is a closed subset of X_j, $1 \leq j \leq n$, then $E_1 \times \cdots \times E_n$ is a closed subset of $X_1 \times \cdots \times X_n$.

2. Prove that the product of a finite number of regular spaces is regular.

3. Prove that the connected components of $X_1 \times \cdots \times X_n$ are the sets of the form $E_1 \times \cdots \times E_n$, where E_j is a connected component of X_j, $1 \leq j \leq n$. Show that a similar result holds for path components.

4. Suppose $X = X_1 \times \cdots \times X_n$, where each X_j is nonempty. Establish the following assertions:

(a) If X is Hausdorff, then each X_j is Hausdorff.

(b) If X is regular, then each X_j is regular.

(c) If X is normal, then each X_j is normal.

(d) If X is connected, then each X_j is connected.

(e) If X is path-connected, then each X_j is path-connected.

(f) If X is compact, then each X_j is compact.

5. Let $\mathcal{T}$ be the half-open interval topology on $\mathbb{R}$, introduced in Exercise 4.6. Let $\mathcal{S}$ be the product topology for $(\mathbb{R}, \mathcal{T}) \times (\mathbb{R}, \mathcal{T})$, so that a base for open sets of $(\mathbb{R}^2, \mathcal{S})$ is the family of sets of the form $[x, x+\varepsilon) \times [y, y+\varepsilon)$, where $x, y \in \mathbb{R}$ and $\varepsilon > 0$.

(a) Prove that the line $L = \{(x, -x) : x \in \mathbb{R}\}$ is $\mathcal{S}$-closed.

(b) Identify the relative topology on L inherited from $\mathcal{S}$.

(c) Let S be a sequence in $\mathbb{R}$ that is dense in $\mathbb{R}$ (in the metric topology). Define

$$E = \{(x, -x) : x \in S\}, \qquad F = \{(x, -x) : x \in \mathbb{R} \setminus S\}.$$

Prove that E and F are disjoint $\mathcal{S}$-closed subsets of $\mathbb{R}^2$.

(d) Suppose $V \in \mathcal{S}$ contains F. Prove that there exist $\varepsilon > 0$, $b \in S$, and a sequence $\{x_n\}$ in $\mathbb{R} \setminus S$ such that $|x_n - b| \to 0$ and such that $[x_n, x_n + \varepsilon) \times [-x_n, -x_n + \varepsilon) \subseteq V$ for all n. *Hint:* Apply the Baire Category Theorem to $\mathbb{R}$.

(e) Prove that there do not exist disjoint $\mathcal{S}$-open neighborhoods U of E and V of F. In particular, deduce that $(\mathbb{R}, \mathcal{T}) \times (\mathbb{R}, \mathcal{T})$ is not normal.

11. Set Theory and Zorn's Lemma

1. Let $\mathcal{T}$ be a family of subsets of a set S. Suppose that a subset E of S belongs to $\mathcal{T}$ if and only if every finite subset of E belongs to $\mathcal{T}$. Prove that $\mathcal{T}$ has a maximal element.

2. Let R be a commutative ring with an identity. Show that every proper ideal in R is contained in a maximal (proper) ideal in R.

Note: The concepts of algebra occurring in this exercise are not needed elsewhere in this book.

3. Let A be a linearly independent subset of a vector space V. Show that there is a basis B for V such that $B \supseteq A$.

4. Let W be a subspace of a real vector space V. Show that every real-valued linear functional L on W is the restriction to W of a real-valued linear functional $\widetilde{L}$ on V. *Hint:* Take a basis for W and, by Exercise 3, extend it to a basis for V.

5. Let $\mathfrak{X}$ be a normed real vector space. Prove that if $\mathfrak{X}$ is infinite-dimensional, then there is a discontinuous linear functional on $\mathfrak{X}$ (cf. Exercise I.7.5).

6. Let $\mathfrak{X}$ be an infinite-dimensional Banach space. Show that any vector-space basis for $\mathfrak{X}$ is uncountable. *Hint:* Suppose $\{x_1, x_2, \ldots\}$ is a basis for $\mathfrak{X}$. Define E_m to be the set of linear combinations of $\{x_1, \ldots, x_m\}$ and apply the Baire Category Theorem.

12. Infinite Product Spaces

1. Prove that if E_α is a closed subset of X_α for all α, then $\prod E_\alpha$ is a closed subset of $\prod X_\alpha$.

2. Prove that each projection π_β of $\prod X_\alpha$ onto a coordinate space X_β is an open map.

3. Prove that the restriction of the projection π_β to the coordinate slice $X_\beta \times \{y\}$ is a homeomorphism from $X_\beta \times \{y\}$ to X_β, $X_\beta \times \{y\}$ being given the relative topology as a subset of $\prod X_\alpha$.

4. Prove that the product of Hausdorff spaces is Hausdorff.

5. Prove that the product of regular spaces is regular.

6. Prove that the product of connected spaces is connected. *Hint:* Show that a connected component of the product is dense and apply Exercise 8.9.

7. Identify the connected components of $X = \prod X_\alpha$ in terms of those of the X_α's. Prove your answer.

8. Prove that the product of path-connected spaces is path-connected.

9. Let S_α be a nonempty set for each α belonging to an index set A. Using Tychonoff's Theorem and elementary reasoning, prove that $\prod S_\alpha$ is not empty. Deduce that Tychonoff's Theorem implies the Axiom of Choice.

Hint: Let X_α be obtained from S_α by adjoining one point p_α. Endow X_α with the topology whose open sets are the cofinite subsets of X_α together with the empty set and the singleton $\{p_\alpha\}$. Show that the subsets $\pi_\alpha^{-1}(S_\alpha)$ of $\prod X_\alpha$ have the finite intersection property.

10. Let S be a set. For each $s \in S$, let X_s be the discrete topological space consisting of two points $\{0, 1\}$ and let $X = \prod\{X_s : s \in S\}$.

(a) Show that X may be identified with the family of all subsets of S.

(b) Prove that X is a compact Hausdorff space.

(c) When is X second-countable?

(d) When does X satisfy the first axiom of countability (defined in Exercise 4.4)?

11. Let X_α, $\alpha \in A$, be a collection of topological spaces. Let $\mathcal{B}$ be the family of subsets of $\prod X_\alpha$ of the form $\prod U_\alpha$, where each U_α is an open subset of X_α.

(a) Show that $\mathcal{B}$ forms a base for a topology for $\prod X_\alpha$. (This topology is called the *box topology.*)

(b) Show that if each X_α is discrete, then the box topology is the discrete topology for $\prod X_\alpha$.

(c) Show that $\prod X_\alpha$ with the box topology, need not be compact, even though each X_α is compact. *Hint:* Let each X_α be the discrete space consisting of two points.

(d) Show that if each X_α is Hausdorff, then the box topology for $\prod X_\alpha$ is a Hausdorff topology. Is the same assertion true for regularity? for normality? Justify your answers.

12. Let A be an index set, and for each $\alpha \in A$, let $X_\alpha = \mathbb{R}$.

(a) Let w be any real-valued function on A such that $w(\alpha) > 0$ for all $\alpha \in A$. Fix $x \in \prod X_\alpha$ and let E be the set of $y \in \prod X_\alpha$ such that $|x(\alpha) - y(\alpha)|w(\alpha)$ is bounded on A. Prove that E is a closed and open subset of $\prod X_\alpha$, with the box topology from Exercise 11.

(b) If the index set A is infinite, prove that $\prod X_\alpha$, with the box topology, is disconnected.

13. Quotient Spaces

1. Let $X/\sim$ be the quotient space determined by an equivalence relation $\sim$ on a topological space X. Prove the following assertions:

(a) If X is compact, then $X/\sim$ is compact.

(b) If X is connected, then $X/\sim$ is connected.

(c) If X is path-connected, then $X/\sim$ is path-connected.

2. Let $\sim$ be an equivalence relation on a topological space X. Prove that $X/\sim$ is a T_1-space if and only if each equivalence class is closed. Give an example of a T_1-space X and an equivalence relation $\sim$ such that $X/\sim$ is not a T_1-space.

3. Define an equivalence relation in $X = [0,1] \times [0,1]$ by declaring $(s_0, t_0) \sim (s_1, t_1)$ if and only if $t_0 = t_1 > 0$. Describe the quotient space $X/\sim$ and show that it is not a Hausdorff space.

4. Let f be an open continuous map of a topological space X onto a topological space Y. Show that Y is homeomorphic to the quotient space of X obtained by identifying each level set of f to a point.

5. Let E be a closed subset of a compact Hausdorff space X. Prove that the quotient space obtained from X by identifying E to a point is homeomorphic to the one-point compactification of $X \setminus E$.

6. Let B^n be the closed unit ball in $\mathbb{R}^n$. Prove that the quotient space obtained from B^n by identifying its boundary S^{n-1} to a point is homeomorphic to the n-sphere S^n.

7. Let $X = X_1 \times \cdots \times X_n$ be a product of topological spaces. Define an equivalence relation $\sim$ on X by declaring $(x_1, \ldots, x_n) \sim (y_1, \ldots, y_n)$ if and only if $x_1 = y_1$. Show that $X/\sim$ is homeomorphic to X_1. Prove an analogous result for an infinite product space $X = \prod_{\alpha \in A} X_\alpha$.

8. For $n \geq 1$, define $P^n = S^n/\sim$, where the equivalence relation is defined by declaring $x \sim y$ if and only if $x = y$ or $x = -y$. In other words, P^n is obtained from S^n by identifying pairs of antipodal points. The space P^n is called *real projective space* of dimension n, and it can be regarded as the set of lines in $\mathbb{R}^{n+1}$ which pass through the origin. Establish the following assertions:

(a) P^n is a compact Hausdorff space.

(b) The projection $\pi : S^n \to P^n$ is a local homeomorphism, that is, each $x \in S^n$ has an open neighborhood that is mapped homeomorphically by π onto an open neighborhood of $\pi(x)$.

(c) P^1 is homeomorphic to the circle S^1.

(d) P^n is homeomorphic to the quotient space obtained from the closed unit ball B^n in $\mathbb{R}^n$ by identifying antipodal points of its boundary S^{n-1}.

9. Let $n \geq 1$ and let $X = \mathbb{C}^{n+1} \setminus \{0\}$. Let $\sim$ be the equivalence relation in X obtained by declaring $x \sim y$ if $x = \lambda y$ for some complex number λ. Then $CP^n = X/\sim$ is called *complex projective space* of dimension n. It can be regarded as the set of one-dimensional subspaces of the complex vector space $\mathbb{C}^{n+1}$.

(a) Prove that CP^n is a compact Hausdorff space.

(b) Prove that CP^1 is homeomorphic to S^2.

(c) Regard S^{2n+1} as the set of vectors $z = (z_0, \ldots, z_n) \in \mathbb{R}^{n+1}$ such that $|z_0|^2 + \cdots + |z_n|^2 = 1$. Show that the natural projection $\pi : S^{2n+1} \to CP^n$ is onto and that $\pi^{-1}(x)$ is homeomorphic to S^1 for all $x \in CP^n$. Show that each $x \in CP^n$ has an open neighborhood U such that $\pi^{-1}(U)$ is homeomorphic to the product space $U \times S^1$.

Remark: Intuitively speaking, S^{2n+1} is a "twisted product" of CP^n and S^1. We say that S^{2n+1} is a *fiber space* over CP^n, with *fiber* S^1.

III. Homotopy Theory

1. Groups

1. Prove that any two groups with one element are isomorphic. Prove that any two groups with two elements are isomorphic. Prove that any two groups with three elements are isomorphic.

2. Show that for any fixed positive integer n, there are at most a finite number of nonisomorphic groups of order n. *Hint:* If $a_1, \ldots, a_n$ are the elements of the group, then the group is determined by knowing all the products $a_i * a_j$, $1 \leq i \leq j \leq n$.

3. Find the number of nonisomorphic groups of order 4.

4. Prove that if f is a homomorphism from a group G to a group H, then $f(G)$ is a subgroup of H. Prove that if e is the identity of H, then $f^{-1}(e)$ is a subgroup of G.

5. Let G and H be groups. Define an operation in the Cartesian product $G \times H$ by

$$(a_1, b_1) * (a_2, b_2) = (a_1 * a_2, b_1 * b_2), \qquad a_1, a_2 \in G;\ b_1, b_2 \in H.$$

Show that endowed with this operation, $G \times H$ becomes a group. Show that $G \times H$ is commutative if and only if G and H are commutative.

Remark. $G \times H$ is the *direct product* of G and H. If G and H are abelian and if the operations in G and H are denoted by "+", then it is customary to refer to the direct product of G and H as the *direct sum* of G and H and denote it by $G \oplus H$. For example, the group $\mathbb{R} \oplus \mathbb{R}$ is isomorphic to $\mathbb{R}^2$ and $\mathbb{Z} \oplus \mathbb{Z}$ is isomorphic to the subgroup of integral lattice points in $\mathbb{R}^2$.

2. Homotopic Paths

1. Suppose that $(\alpha\beta)\gamma = \alpha(\beta\gamma)$ for any three paths α, β, and γ in X for which the product is defined. Show that each path component of X consists of a single point.

2. Let X be path-connected and let $b \in X$. Show that every path in X is homotopic with endpoints fixed to a path passing through b.

3. Let D be an open subset in $\mathbb{R}^n$, let α be a path in D from x to y, and set

$$d = \inf\{|\alpha(s) - w| : w \in \partial D, 0 \leq s \leq 1\}.$$

Show that if β is any path in D from x to y such that $|\beta(s) - \alpha(s)| < d$, $0 \leq s \leq 1$, then β is homotopic to α with endpoints fixed.

4. A path α in $\mathbb{R}^n$ is *polygonal* if there is a subdivision $0 = s_0 < s_1 < \cdots < s_m = 1$ of the unit interval such that α maps each interval $[s_{j-1}, s_j]$ onto the straight-line segment from $\alpha(s_{j-1})$ to $\alpha(s_j)$. Show that every path in an open subset D of $\mathbb{R}^n$ is homotopic in D with endpoints fixed to a polygonal path.

5. Show that any path in S^n is homotopic with endpoints fixed to a polygonal path on S^n, where "polygonal" is now interpreted to mean that the path is formed from arcs lying on great circles of S^n.

6. Let (X, d) be a compact metric space and let $a, b \in X$. Let $\mathcal{P}$ be the set of all paths in X from a to b with the metric

$$\rho(\alpha, \beta) = \sup\{d(\alpha(s), \beta(s)) : 0 \leq s \leq 1\}.$$

Show that two paths $\alpha, \beta \in \mathcal{P}$ are homotopic with endpoints fixed if and only if α and β lie in the same path component of $\mathcal{P}$.

3. The Fundamental Group

1. Prove that if $n \geq 2$, then S^n is simply connected. *Hint:* Use Exercise 2.5 to show that every loop in S^n is homotopic to a loop that does not cover all of S^n.

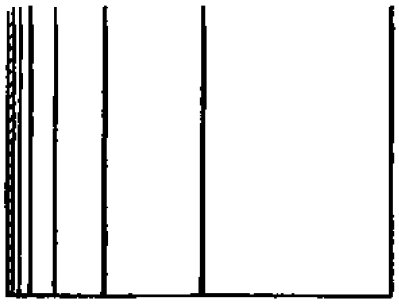
2. Prove that if there are simply connected open subsets U and V of X such that $U \cup V = X$ and $U \cap V$ is nonempty and path-connected, then X is simply connected.

3. A space X is *contractible to a point* $x_0 \in X$ *with* x_0 *held fixed* if there is a map $F : X \times [0,1] \to X$ such that

$$\begin{aligned} F(x,0) &= x_0, \quad x \in X, \\ F(x,1) &= x, \quad x \in X, \\ F(x_0,t) &= x_0, \quad 0 \leq t \leq 1. \end{aligned}$$

Show that such a space is simply connected.

4. Let X be the *comb space*, that is, the compact subset of $\mathbb{R}^2$ consisting of the horizontal interval $\{(x, 0) : 0 \leq x \leq 1\}$ and the closed vertical intervals of unit length with lower endpoints at $(0, 0)$ and at $(0, 1/n)$, $1 \leq n < \infty$.



Comb space

Show that X is contractible to $(0, 0)$ with $(0, 0)$ held fixed. Show that X is not contractible to $(0, 1)$ with $(0, 1)$ held fixed.

5. A subset W of $\mathbb{R}^n$ is *star-shaped* with respect to a point $w \in W$ if, whenever $y \in W$, then the straight-line segment from w to y is contained in W.

(a) Show that a subset W of $\mathbb{R}^n$ is convex if and only if it is star-shaped with respect to each of its points.

(b) Give an example of a star-shaped set that is not convex.

(c) Show that a star-shaped set in $\mathbb{R}^n$ is contractible to a point.

(d) Show that a star-shaped set in $\mathbb{R}^n$ is simply connected.

6. Let (X, x_0) and (Y, y_0) be pointed spaces. Show that $\pi_1(X \times Y, (x_0, y_0))$ is isomorphic to the direct product $\pi_1(X, x_0) \times \pi_1(Y, y_0)$.

7. Prove that the product of simply connected spaces is simply connected.

8. Prove that if $n \geq 3$, then $\mathbb{R}^n \setminus \{0\}$ is simply connected.

9. Let X be a path-connected topological space and let $b, c \in X$. Let B^2 be the closed unit ball in $\mathbb{R}^2$, with boundary circle S^1. Show that the following are equivalent.

(a) X is simply connected.

(b) Any two paths from b to c are homotopic with endpoints fixed.

(c) Every map $f : S^1 \to X$ extends to a map $F : B^2 \to X$.

4. Induced Homomorphisms

1. Show that simple connectivity is a topological property.

2. A subspace A of a topological space X is a *retract* of X if there is a map $f : X \to A$ such that $f(y) = y$ for all $y \in A$. The map f is called a *retraction* of X onto A. Show that the unit sphere S^n in $\mathbb{R}^{n+1}$ is a retract of $\mathbb{R}^{n+1} \setminus \{0\}$.

3. Let f be a retraction of X onto A, and let $x_0 \in A$. Let $j : A \to X$ be the inclusion map. Prove the following:

(a) $j_* : \pi_1(A, x_0) \to \pi_1(X, x_0)$ is one-to-one.

(b) $f_* : \pi_1(X, x_0) \to \pi_1(A, x_0)$ is onto.

(c) If X is simply connected, then A is simply connected.

5. Covering Spaces

1. For m an integer, let α_m be the loop in S^1 defined by

$$\alpha_m(s) = e^{2\pi i m s}, \qquad 0 \leq s \leq 1.$$

Show that every loop in S^1 based at 1 is homotopic with endpoints fixed to precisely one of the loops α_m.

2. Suppose that $p : (E, e) \to (X, b)$ is a covering map and that E is simply connected. Suppose furthermore that E and X are groups with identities e and b, respectively, and that p is a homomorphism. Suppose finally that for each fixed $y \in E$, the group multiplication $z \mapsto y * z$ is a continuous function on E. Prove that the fiber $p^{-1}(b)$ is a subgroup of E, and that $\pi_1(X, b) \cong p^{-1}(b)$.

3. Show that the exponential map $p : \mathbb{R}^n \to T^n$, defined earlier in this section, is a covering map of $\mathbb{R}^n$ onto the n-torus T^n. Show that

$$\pi_1(T^n) \cong \mathbb{Z} \oplus \cdots \oplus \mathbb{Z} \qquad (n \text{ summands}).$$

For each n-tuple $(m_1, \ldots, m_n) \in \mathbb{Z} \oplus \cdots \oplus \mathbb{Z}$, define explicitly a loop in T^n based at $(1, \ldots, 1)$ in the corresponding homotopy class.

4. Show that the map $p : \mathbb{C} \to \mathbb{C} \setminus \{0\}$, defined by

$$p(z) = e^z, \qquad z \in \mathbb{C},$$

is a covering map. What is $\pi_1(\mathbb{C} \setminus \{0\})$?

5. Show that the restriction of the map p of Exercise 4 to the horizontal strip $E = \{x + iy : c < y < d\}$ is a covering map of E over the open annulus $\{w : e^c < |w| < e^d\}$. What is the fundamental group of the open annulus? What is the fundamental group of a closed annulus?

6. Let X be the figure-eight space, and let α and β be the loops in X indicated in the figure and discussion preceding Theorem 5.4. Show that every element of $\pi_1(X)$ can be expressed uniquely as a finite product

$$[\alpha]^{m_1}[\beta]^{m_2}[\alpha]^{m_3}\cdots,$$

where $m_1, m_2, \ldots$ are integers and $m_j \neq 0$ for $j \geq 2$ ($m_1 = 0$ is allowed).

Note: This proves Van Kampen's Theorem, that $\pi_1(X)$ is the free group with generators $[\alpha]$ and $[\beta]$. For the uniqueness assertion, construct an appropriate covering space of X.

7. A topological space Y is *locally path-connected* if for each $y \in Y$ and neighborhood U of y, there exists a neighborhood V of y such that every point of V can be joined to y by a path in U. Let (Y, c) be a pointed topological space such that Y is locally path-connected and simply connected, let $p : (E, e) \to (X, b)$ be a covering map.

(a) Show that every map $f : (Y, c) \to (X, b)$ can be uniquely lifted to a map $g : (Y, c) \to (E, b)$.

(b) Suppose in addition that X is locally path-connected and E is simply connected. Show that if f is a covering map, then g is a homeomorphism of Y and E. (In other words, a simply connected covering space of a locally path-connected space is unique.)

8. Let $p : E \to X$ be a covering map. A map $f : E \to E$ is a *covering transformation* if $p \circ f = f$.

(a) Show that with the operation of composition, the covering transformations form a group.

(b) Show that if X is locally path-connected and E is simply connected, then the group of covering transformations is isomorphic to the fundamental group of X.

9. What are the covering transformations of S^2 over P^2? of $\mathbb{R}$ over S^1?

10. Let X be a path-connected topological space such that every $x \in X$ has an open neighborhood that is simply connected. Fix $b \in X$ and let E be the set of all pairs $(x, [\gamma])$, where $x \in X$ and γ is a path in X from b to x. For each simply connected open subset U of X and each path γ in X starting at b and terminating at some point $\gamma(1) \in U$, define $W(U, \gamma)$ to be the set of all pairs $(x, [\gamma][\alpha])$ in E such that $x \in U$ and α is a path in U from $\gamma(1)$ to x. Prove the following.

(a) The sets $W(U, \gamma)$ form a base for a topology for E.

(b) The natural projection $(x, [\gamma]) \mapsto x$ of E onto X is a covering map.

(c) E is simply connected.

Remark: By Exercise 7, the space E is essentially unique. It is called the *universal covering space* of X.

11. Let X be the quotient space obtained from the union of circles $\{x^2 + y^2 = 1, z = 1/n\}$ in $\mathbb{R}^3$, for $1 \leq n \leq \infty$, by identifying the set $\{(1, 0, 1/n) : 1 \leq n \leq \infty\}$ to a point b. Show that if $p : (E, e) \to (X, b)$ is a covering map, then E is not simply connected. (Thus X has no universal covering space. Why does Exercise 10 not apply?)

6. Some Applications of the Index

1. Prove in detail that the function g appearing in the proof of the Brouwer Fixed Point Theorem is continuous.

2. Prove in detail that the functions g_0, g_1, and g_2 appearing in the proof of the Ham Sandwich Theorem are continuous.

3. Let U, V, and W be open balls in $\mathbb{R}^3$. Describe the plane in $\mathbb{R}^3$ which divides each of U, V, and W in half by volume. When is the plane unique?

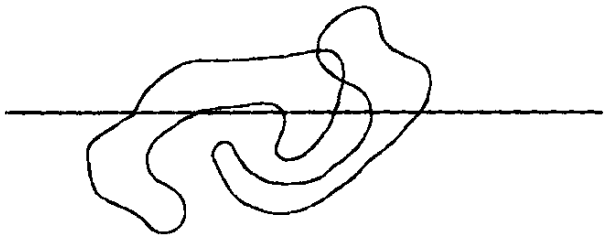
4. If U and V are bounded connected open subsets of $\mathbb{R}^3$ and $w \in \mathbb{R}^3$, then prove that there exists a plane in $\mathbb{R}^3$ passing through w and dividing U and V in half by volume.

5. A topological space X has the *fixed-point property* if every map from X to X has a fixed point. Show that the fixed-point property is a topological property.

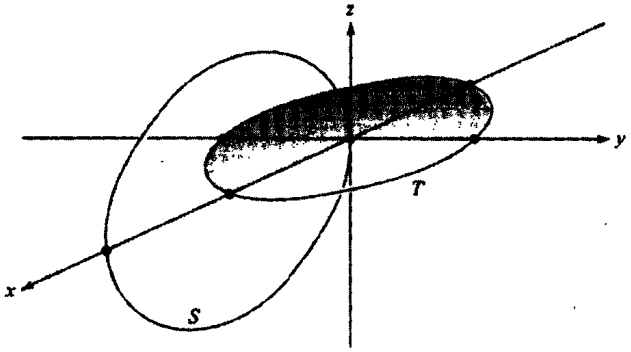
6. Recall (from Exercise 4.2) that a subspace Y of X is a *retract* of X if there is a map $f : X \to Y$ such that f is the identity on Y. Show that the unit ball B^n in $\mathbb{R}^n$ has the fixed-point property if and only if its boundary S^{n-1} is not a retract of B^n. (It turns out that B^n has the fixed-point property. See Section IV.2.)

7. Prove that if f is a map from S^1 to $\mathbb{R}$, then there exist antipodal points of S^1 at which f has equal values.

8. Prove that if U and V are bounded connected open subsets of $\mathbb{R}^2$, then there exists a straight line that divides each of U and V in half by area. (This is the *Pancake Theorem*.)



9. Let S be the circle in $\mathbb{R}^3$ centered at $(1,0,0)$, with radius 1, and contained in the plane $\{y=0\}$. Let T be the circle in $\mathbb{R}^3$ centered at $(0,0,0)$, with radius 1, and contained in the plane $\{z=0\}$. Show that S and T are linked, that is, there does not exist a map $f : B^2 \to \mathbb{R}^3 \setminus T$ such that the restriction of f to S^1 maps S^1 homeomorphically onto S. (Intuitively, every membrane with boundary S must be punctured somewhere by T.) *Hint:* Show first that S is a retract of $\mathbb{R}^3 \setminus T$. A retraction is given by first projecting $\mathbb{R}^3 \setminus T$ circularly onto the punctured half-space $\{(x,0,z) : x \geq 0\} \setminus (1,0,0)$ and then projecting radially onto the circle S.



7. Homotopic Maps

1. Prove in detail that the relation of being homotopically equivalent is transitive.

2. Show that if Y is contractible, then $X \times Y$ is homotopically equivalent to X.

3. Prove that any annulus in $\mathbb{R}^2$ is homotopically equivalent to S^1.

4. Prove that the punctured open disc $\{z : 0 < |z| < 1\}$ in $\mathbb{C}$ is homotopically equivalent to S^1.

5. Prove that $\mathbb{R}^{n+1} \setminus \{0\}$ is homotopically equivalent to S^n.

6. Prove that the doubly punctured plane is homotopically equivalent to the figure eight.

7. Prove that if Y is contractible, then any two maps from X to Y are homotopic.

8. Prove that if X is contractible and Y is path-connected, then any two maps from X to Y are homotopic.

9. Let (X, b) be a pointed topological space, let α_0 and α_1 be loops in X based at b, and let $f_j(e^{2\pi is}) = \alpha_j(s)$, $0 \leq s \leq 1$, be the corresponding maps of S^1 into X. Show that f_0 is homotopic to f_1 if and only if there exists a loop β in X based at b such that

$$[\alpha_1] = [\beta]^{-1}[\alpha_0][\beta].$$

10. If X is path-connected and $\pi_1(X)$ is commutative, show that there is a one-to-one correspondence between elements of $\pi_1(X)$ and homotopy classes of maps of S^1 into X.

8. Maps into the Punctured Plane

1. Let X be compact and let $f : X \to \mathbb{C} \setminus \{0\}$ be an exponential. Show that there exists $\varepsilon > 0$ such that every map $g : X \to \mathbb{C} \setminus \{0\}$ which satisfies $|f(x) - g(x)| < \varepsilon$, $x \in X$, is an exponential.

2. Let X be a locally compact Hausdorff space. Show that two maps f and g from X to $\mathbb{C} \setminus \{0\}$ are homotopic if and only if f/g is an exponential. *Hint:* Consider first the case in which X is compact.

3. Let X be a locally compact Hausdorff space that is contractible. Show that any map from X to $\mathbb{C} \setminus \{0\}$ is an exponential.

4. Let f be a map from S^1 to $\mathbb{C} \setminus \{0\}$. Show that there exists $\varepsilon > 0$ such that any map $g : S^1 \to \mathbb{C} \setminus \{0\}$ satisfying $|g(z) - f(z)| < \varepsilon$, $z \in S^1$, has the same index as f.

5. Suppose that f and g are maps from S^1 to S^1 such that f and g do not assume antipodal values at any point of S^1. Show that $\text{ind}(f) = \text{ind}(g)$.

6. If $n \geq 2$, show that every map $f : S^n \to \mathbb{C} \setminus \{0\}$ is an exponential.

7. If $n \geq 2$, show that every map $f : P^n \to \mathbb{C} \setminus \{0\}$ is an exponential. (Note, however, that P^n is not simply connected: $\pi_1(P^n) = \mathbb{Z}_2$ for $n \geq 2$. Can you offer an explanation for this state of affairs?)

8. Classify the homotopy classes of maps from the figure eight to $\mathbb{C} \setminus \{0\}$.

9. Vector Fields

1. Sketch a vector field on S^2 that has precisely one zero.

2. Show that the only constant vector field tangent to S^2 is identically zero.

3. Let F be a continuous function from S^2 to $\mathbb{R}^3$. Show that there is a point $p \in S^2$ at which $F(p)$ is normal to S^2, that is, $F(p)$ is a multiple of p.

4. Let φ be a continuously differentiable map from $\mathbb{R}^n$ to $\mathbb{R}^m$. Then φ induces a transformation φ^* from vector fields on subsets of $\mathbb{R}^n$ to vector fields on $\mathbb{R}^m$ as follows. For a vector field $F = (F_1, \ldots, F_n)$ on $\mathbb{R}^n$, define $\varphi^* F = ((\varphi^* F)_1, \ldots, (\varphi^* F)_m)$ on $\mathbb{R}^m$ by the formula

$$(\varphi^* F)_j(\varphi(p)) = \sum_{k=1}^{n} \frac{\partial \varphi_j}{\partial x_k}(p) F_k(p), \qquad 1 \leq j \leq m.$$

Show that if $\sigma : S^2 \setminus N \to \mathbb{R}^2$ is the stereographic projection, if $F = (F_1, F_2, F_3)$ is a vector field on S^2, and if G is the vector field on $\mathbb{R}^2$ defined by (9.2), then

$$(\sigma^* F)(u, v) = (1 - z)^{-2} G(u, v),$$

where $(u, v) = \sigma(x, y, z)$. In particular, $\sigma^* F$ and G point in the same direction.

5. A *closed curve* (= closed path = loop) in the complex plane $\mathbb{C}$ is a continuous mapping γ of S^1 into $\mathbb{C}$. If $p \notin \gamma(S^1)$, then the *winding number* of the closed curve γ around p is the index of the map $e^{i\theta} \mapsto \gamma(e^{i\theta}) - p$, $0 \leq \theta \leq 2\pi$. Show that the winding numbers of γ around p are identical for all p in a fixed connected component of $\mathbb{C} \setminus \gamma(S^1)$. Show that the winding number of γ around p is zero for p in the unbounded connected component of $\mathbb{C} \setminus \gamma(S^1)$.

6. Suppose that $F : U \to \mathbb{R}^2$ is a continuous vector field on an open subset U of $\mathbb{R}^2$. Let p be an isolated zero of F, i.e., suppose that there is a disc $\overline{B(p;\varepsilon)} = \{q : |q-p| \leq \varepsilon\}$ contained in U such that $F(q) \neq 0$ for $q \in \overline{B(p;\varepsilon)}$, $q \neq p$, while $F(p) = 0$.

(a) Show that the index of F around γ is the same for any loop γ in $\overline{B(p;\varepsilon)} \setminus \{p\}$ with winding number 1 around p. This common value is called the *index of F at p*.

(b) Express the index of F around γ in terms of the index of F at p, when γ is a loop in $\overline{B(p;\varepsilon)}\setminus\{p\}$ with winding number m around p. Prove your answer.

(c) What are the indices at the origin of the vector fields depicted in the figures discussed at the beginning of this section?

(d) If F has index m at an isolated zero p, what is the index of $-F$ at p?

7. Suppose that $F : \overline{B(0;1)} \to \mathbb{R}^2$ is a continuous vector field on the closed unit disc $\overline{B(0;1)} = \{(x,y) : x^2 + y^2 \leq 1\}$ in $\mathbb{R}^2$ with only a finite number of zeros, all of which lie in the open disc $B(0;1)$. Show that the index of F around the positively traversed unit circle is the sum of the indices of F at its zeros. *Hint:* Suppose first that F has only two zeros, say at $(-\frac{1}{2}, 0)$ and $(\frac{1}{2}, 0)$. Show that the index of F around the unit circle is the sum of the indices of F around appropriate semicircular loops separating the zeros.

8. Let $F : S^2 \to \mathbb{R}^2$ be a vector field on the sphere with only a finite number of zeros, such that $F \neq 0$ near the north pole N. Let $\sigma : S^2 \setminus \{N\} \to \mathbb{R}^2$ be the stereographic projection. Show that the index of $F \circ \sigma^{-1}$ around a large circular loop $e^{i\theta} \mapsto (R\cos\theta, R\sin\theta)$, $0 \leq \theta \leq 2\pi$, is $+2$. (Thus with reasonable definitions, the sum of the indices of a vector field on S^2, with only isolated zeros, is $+2$. Check this for the vector fields depicted in the figures in the discussion after Lemma 9.1.)

10. The Jordan Curve Theorem

1. Let X be a connected, locally path-connected space and let U and V be simply connected open subsets of X which cover X. Show that if $U \cap V$ consists of two connected components with disjoint closures, then $\pi_1(X) \cong \mathbb{Z}$.

2. Let $h : [0, \infty) \to \mathbb{R}^2$ be a one-to-one continuous map such that $|h(t)| \to \infty$ as $t \to \infty$. Show that $\mathbb{R}^2 \setminus h([0, \infty))$ is connected. *Hint:* Follow the proof of Theorem 10.3 and use Exercise 1.

3. A *simple arc* in a space X is the homeomorphic image in X of the unit interval $[0, 1]$. Show that the complement of a simple arc in $\mathbb{R}^2$ or in S^2 is connected.

4. If γ is a simple closed curve in $\mathbb{R}^2$ with image Γ, then show that each component of $\mathbb{R}^2 \setminus \Gamma$ has boundary Γ. *Hint:* Use Exercise 3 or Exercise 6.

5. Let E be a compact subset of the complex plane $\mathbb{C}$ and let $p \in \mathbb{C} \setminus E$. Prove that p lies in the unbounded connected component of $\mathbb{C} \setminus E$ if and only if $z - p$ is an exponential on E.

6. Using Exercise 5, show that if E is a compact contractible subset of $\mathbb{C}$, then $\mathbb{C} \setminus E$ is connected. (This generalizes Exercise 3.)

7. Let E be a compact subset of $\mathbb{C}$ and suppose $p_1, \ldots, p_m$ lie in different bounded components of $\mathbb{C} \setminus E$. Let $q_1, \ldots, q_m$ be integers. Prove that the function

$$f = (z - p_1)^{q_1} \cdots (z - p_m)^{q_m}$$

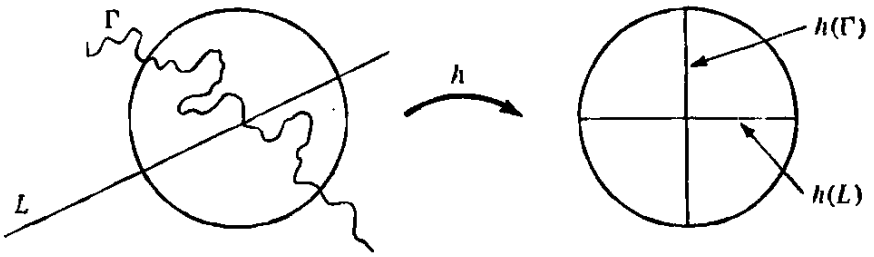
is an exponential on E if and only if $q_1 = \cdots = q_m = 0$. *Hint:* Let U be the connected component of $\mathbb{C} \setminus E$ containing p_1. Write $f = e^h$ on E, extend h continuously to $\mathbb{C}$, and consider the function g defined to be f on U and e^h on $\mathbb{C} \setminus U$.

8. Using Exercise 7, show that the complement of a simple closed curve in $\mathbb{C}$ has at most two connected components.

9. Using Exercises 4, 6, and 8, show that the Jordan Curve Theorem is valid, providing that Γ contains a straight-line segment. *Hint:* By considering the index of $\gamma - p$, show that $z - p$ cannot be an exponential for p lying on both sides of the straight-line segment in Γ.

10. Prove the Jordan Curve Theorem by introducing a straight line segment in $\mathbb{C} \setminus \Gamma$ between two appropriate points of Γ and applying Exercise 9 to the resulting simple closed curves.

11. A straight line L in $\mathbb{R}^2$ is *topologically transversal* to a simple closed curve Γ in $\mathbb{R}^2$ if for every $p \in \Gamma \cap L$, there is an $\varepsilon > 0$ and a homeomorphism of $B(p;\varepsilon)$ onto $B((0,0);1)$ mapping $B(p;\varepsilon) \cap \Gamma$ onto the vertical interval $\{(0,t) : -1 < t < 1\}$ and mapping $B(p;\varepsilon) \cap L$ onto the horizontal interval $\{(s,0) : -1 < s < 1\}$. Informally, Γ can be straightened out near p without bending L.



(a) Prove that if L is topologically transversal to Γ, then the number of points in $\Gamma \cap L$ is finite and even. *Hint:* Use Exercise 4.

(b) Prove that if $q \in L \setminus \Gamma$, then q is in the bounded component of $\mathbb{R}^2 \setminus \Gamma$ if and only if the number of points in $L \cap \Gamma$ on each side of q is odd.

(c) Formulate and prove a similar result for a closed half-line.

IV. Higher Dimensional Homotopy

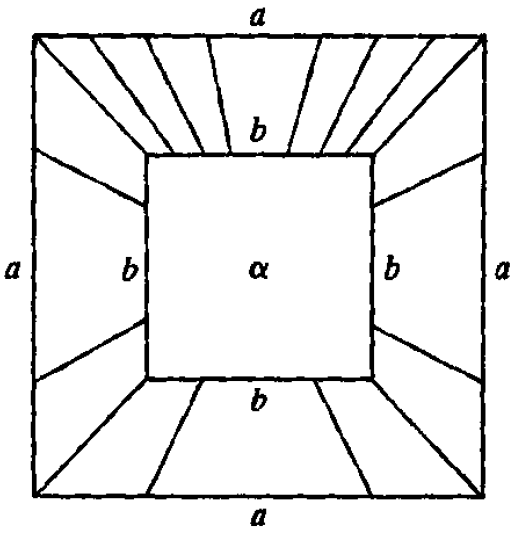
1. Higher Homotopy Groups

1. Prove that $\pi_k(S^1, 1) = 0$ if $k > 1$.

2. Prove that $\pi_k(X \times Y, (b, c)) \cong \pi_k(X, b) \times \pi_k(Y, c)$, $k \geq 1$.

3. Prove in detail that the multiplication defined on $\pi_k(X, b)$ is associative.

4. Let $a, b \in X$. Show that a path γ from a to b induces in a "natural" way an isomorphism of $\pi_k(X, b)$ and $\pi_k(X, a)$. Prove that if X is simply connected, this isomorphism is independent of the path γ. *Hint:* To the k-cube α at b in X, assign a k-cube $\overline{\alpha}$ at a in X, as indicated by the accompanying figure, where the lines from the a-rim to the b-rim are to be each mapped to X by γ.



5. Let $f : S^k \to X$ satisfy $f(p) = b$ and suppose f is homotopic to a constant map. Show that f is homotopic to the constant map at b through a homotopy which fixes p, that is, that $[f]$ is the identity of $\pi_k(X, b)$.

6. Show that if X is contractible to a point b, then $\pi_k(X, b) = 0$ for all $k \geq 1$.

7. Show that a map $g : (Y, c) \to (X, b)$ induces in a "natural" way homomorphisms $g_* : \pi_k(Y, c) \to \pi_k(X, b)$, $k \geq 1$. Show that if $h : (W, a) \to (Y, c)$ is another map, then $(g \circ h)_* = g_* \circ h_*$.

8. Prove that if $g : (Y, c) \to (X, b)$ is a homotopy equivalence, then g_* is an isomorphism of $\pi_k(Y, c)$ and $\pi_k(X, b)$, $k \geq 1$.

2. Noncontractibility of S^n

1. Let $x, y \in B^n$, $x \neq y$. Show that the ray in $\mathbb{R}^n$ issuing from x and passing through y meets S^{n-1} at a unique point $h(x, y)$. Derive an explicit formula for $h(x, y)$. Deduce that the point $h(x, y)$ depends continuously on $x, y \in B^n$, $x \neq y$.

2. (a) Show that the function

$$\chi(t) = \begin{cases} \exp(-1/t^2), & t > 0, \\ 0, & t \leq 0, \end{cases}$$

is smooth (infinitely differentiable).

(b) Show that there is a smooth function Ψ on $\mathbb{R}$ such that $\Psi \geq 0$, $\Psi = 0$ outside the closed interval $[0, 1]$, but Ψ is not identically zero. *Hint:* Consider $\chi(t)\chi(2 - t)$.

(c) Show that there is a smooth function φ on $\mathbb{R}$ such that $0 \leq \varphi \leq 1$, $\varphi(s) = 0$ for $s \leq 0$, and $\varphi(s) = 1$ for $s \geq 1$. *Hint:* Consider the indefinite integral of Ψ.

3. Let X be compact. The *cone over* X is the topological space Λ obtained from the product space $X \times [0,1]$ by identifying the slice $X \times \{1\}$ to a point. Prove that X is contractible if and only if there is a retraction of Λ onto the "slice" $X \times \{0\}$.

4. Show by elementary reasoning that if Theorems 2.1 to 2.3 are valid for an integer n, then they are valid for all integers less than n.

5. Prove Theorem 2.2 by using Lemma 2.4 and carrying out in detail the argument outlined as follows. For each smooth map $f = (f_1, \ldots, f_{n+1})$ from B^{n+1} to $\mathbb{R}^{n+1}$, define V_f to be the integral over B^{n+1} of the Jacobian determinant of f:

$$V_f = \int \cdots \int_{B^{n+1}} \det\left(\frac{\partial f_j}{\partial x_k}\right) dx_1 \cdots dx_{n+1}.$$

Let g be another smooth map such that $g = f$ on S^n. Define $F(t,x) = tf(x) + (1-t)g(x)$ and $V(t) = tV_f + (1-t)V_g$, $0 \le t \le 1$. Then

$$\frac{dV}{dt} = \int \cdots \int_{B^{n+1}} \frac{\partial}{\partial t} \det\left(\frac{\partial F_j}{\partial x_k}\right) dx_1 \cdots dx_{n+1}.$$

If Q_l is the matrix obtained from the Jacobian matrix of F by replacing the lth column by the partial derivatives $\partial F_j/\partial t$, $1 \le j \le n+1$, then

$$\det Q_l = \begin{vmatrix} \dfrac{\partial F_1}{\partial x_l \partial x_k} & \dfrac{\partial F_1}{\partial x_2} & \cdots & \dfrac{\partial F_1}{\partial t} & \cdots & \dfrac{\partial F_1}{\partial x_{n+1}} \\ \vdots & \vdots & & \vdots & & \vdots \\ \dfrac{\partial F_{n+1}}{\partial x_l \partial x_k} & \dfrac{\partial F_{n+1}}{\partial x_2} & \cdots & \dfrac{\partial F_{n+1}}{\partial t} & \cdots & \dfrac{\partial F_{n+1}}{\partial x_{n+1}} \end{vmatrix} + \cdots$$

$$+ \begin{vmatrix} \dfrac{\partial F_1}{\partial x_1} & \cdots & \dfrac{\partial F_1}{\partial x_l \partial t} & \cdots & \dfrac{\partial F_1}{\partial x_n} \\ \vdots & & \vdots & & \vdots \\ \dfrac{\partial F_{n+1}}{\partial x_1} & \cdots & \dfrac{\partial F_{n+1}}{\partial x_l \partial t} & \cdots & \dfrac{\partial F_{n+1}}{\partial x_{n+1}} \end{vmatrix} + \cdots$$

$$+ \begin{vmatrix} \dfrac{\partial F_1}{\partial x_1} & \cdots & \dfrac{\partial F_1}{\partial t} & \cdots & \dfrac{\partial^2 F_1}{\partial x_{n+1} \partial x_l} \\ \vdots & & \vdots & & \vdots \\ \dfrac{\partial F_{n+1}}{\partial x_1} & \cdots & \dfrac{\partial F_{n+1}}{\partial t} & \cdots & \dfrac{\partial^2 F_{n+1}}{\partial x_{n+1} \partial x_l} \end{vmatrix}.$$

On account of cancellation by pairs,

$$\frac{\partial}{\partial t} \det\left(\frac{\partial F_j}{\partial x_k}\right) = \sum_{l=1}^{n+1} \frac{\partial}{\partial x_l} \det Q_l.$$

Since $\partial F_j/\partial t\,(t,x) = 0$ if $x \in S^n$, also $\det Q_l(t,x) = 0$ if $x \in S^n$, and consequently

$$\int \cdots \int_{B^{n+1}} \frac{\partial}{\partial x_l} \det Q_l \, dx_1 \cdots dx_{n+1} = 0.$$

It follows that $dV/dt = 0$ and $V_f = V_g$. Now if g is the identity map $g(x) = x$, $x \in B^{n+1}$, then $V_g > 0$, whereas if $f(B^{n+1}) \subseteq S^n$, then the Jacobian determinant of f vanishes and $V_f = 0$. Thus no such f can coincide with the identity on S^n.

6. (For readers familiar with differential forms on manifolds.) Using differential forms, show that if M is a compact oriented manifold with boundary ∂M, then there is no smooth retraction of M onto ∂M. *Hint:* Let α be an $(n-1)$-form on ∂M such that $\int_{\partial M} \alpha > 0$ and let $\beta = F^*(\alpha)$, an $(n-1)$-form on M that coincides with α on ∂M. Apply Stokes' Theorem to $\int_{\partial M} \beta$.

3. Simplexes and Barycentric Subdivision

1. Show that any k-simplex is homeomorphic to the unit ball B^k in $\mathbb{R}^k$ in such a way that the union of the $(k-1)$-faces corresponds to S^{k-1}.

2. Let S be a simplex in $\mathbb{R}^n$ and let f be a linear functional on $\mathbb{R}^n$ whose maximum value on S is c. Show that $f^{-1}(c)$ is a face of S and that furthermore any face of S can be obtained in such a way, as the maximum set of a linear functional on S.

3. How many simplexes are there in the Nth barycentric subdivision of a k-simplex?

4. Show that the intersection of any two faces of simplexes in $\mathfrak{B}^{(N)}$, if not empty, is the simplex generated by their common vertices.

5. Show that the intersection of a face F of a simplex with a simplex T in its Nth barycentric subdivision, if not empty, is the face of T generated by the vertices of T that lie in F.

6. A subset S of a vector space V is *convex* if S includes the line segment between any two of its points, that is, if whenever $x, y \in S$ and $0 < t < 1$, then $tx + (1-t)y \in S$. A *convex combination* of vectors in a set S is a finite sum of the form $\sum t_j x_j$, where $x_j \in S$, $t_j \geq 0$, and $\sum t_j = 1$. Show that S is convex if and only if any convex combination of vectors in S again belongs to S.

7. Let S be a subset of a vector space V. Show that the set $\mathrm{co}(S)$ of convex combinations of elements of S is a convex subset of V, which is the smallest convex subset of V containing S. The set $\mathrm{co}(S)$ is called the *convex hull* of S.

8. Let S be a subset of $\mathbb{R}^n$ such that every element of $\text{co}(S)$ has a unique representation as a convex combination of elements of S. Show that S is an affinely independent set and that $\text{co}(S)$ is a simplex with vertices S.

4. Approximation by Piecewise Linear Maps

1. Prove that a map h from $\mathbb{R}^n$ to $\mathbb{R}^m$ is affine if and only if h preserves convex combinations.

2. Prove that any two k-simplexes are affinely homeomorphic.

3. Let V be a vector space. A *flat of dimension* k, or a k-flat, is a subset of V of the form $x_0 + W$, where $x_0 \in V$ is fixed and W is a k-dimensional subspace of V.

(a) Show that the maximum number of affinely independent vectors in a k-flat is $k+1$.

(b) Show that each set of $k+1$ affinely independent vectors $w_0, \ldots, w_k$ is contained in a unique k-flat, consisting of vectors of the form $w_0 + \sum_{j=1}^{k} s_j(w_j - w_0)$, $s_1, \ldots, s_k \in \mathbb{R}$.

4. Let $g : \mathbb{R}^n \to \mathbb{R}^m$ be affine. Prove the following assertions:

(a) If E is a flat in $\mathbb{R}^n$, then $g(E)$ is a flat in $\mathbb{R}^m$.

(b) If F is a flat in $\mathbb{R}^m$, then $g^{-1}(F)$ is a flat in $\mathbb{R}^n$.

5. Let S be a k-simplex in $\mathbb{R}^n$ and let $w_0, \ldots, w_q \in S$ be affinely independent. Show that the q-flat generated by $w_0, \ldots, w_q$ meets some $(k-q)$-face of S.

6. Let S be an n-simplex in $\mathbb{R}^n$ and let E be a k-flat in $\mathbb{R}^n$ that passes through the interior of S. Show that the intersection of E and any k-face of S is a nowhere-dense subset of the k-face.

7. The following sequence of exercises establishes the fact that the dimension corresponding to a nonempty open subset of Euclidean space is a topological property.

Prove that if f and g are maps from S^n to S^n such that $|f(x) - g(x)| < 2$ for all $x \in S^n$, then f and g are homotopic.

Hint: Observe first that $tf(x) + (1-t)g(x) \neq 0$ for $0 \leq t \leq 1$, $x \in S^n$.

8. Prove that $\pi_k(S^n) = 0$ if $1 < k \leq n$. *Hint:* Using the technique of the proof of Theorem 4.2, show that any map $f : S^k \to S^n$ can be approximated by a map $g : S^k \to S^n$ such that the range of g is a proper subset of S^n.

9. Prove that S^m is not homeomorphic to S^n if $m \neq n$.

10. Prove that $\mathbb{R}^n$ is not homeomorphic to $\mathbb{R}^m$ if $n \neq m$. *Hint:* If X and Y are homeomorphic, then the one-point compactifications of X and Y are also homeomorphic.

11. Prove that $\pi_n(\mathbb{R}^{n+1} \setminus \{0\}) \neq 0$ for $n \geq 1$.

12. Prove that if U is an open subset of $\mathbb{R}^{n+1}$ and if $p \in U$, then $\pi_n(U \setminus \{p\}) \neq 0$.

13. Prove that if B is a ball in $\mathbb{R}^{n+1}$ and if $p \in B$, then $\pi_k(B \setminus \{p\}) = 0$ for $1 \leq k < n$.

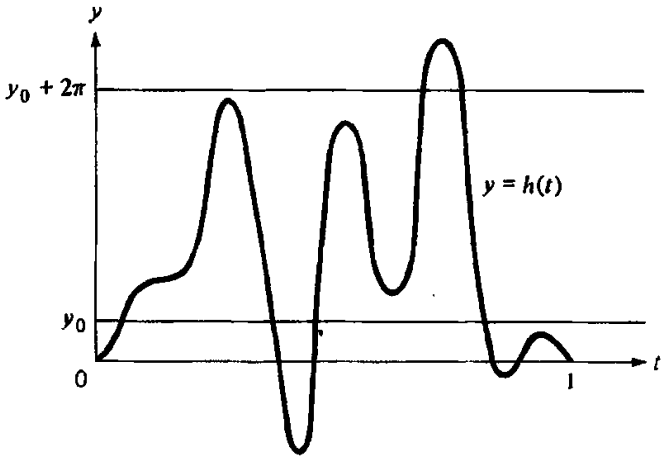
14. Prove that if $n \neq m$, then no nonempty open subset of $\mathbb{R}^n$ is homeomorphic to an open subset of $\mathbb{R}^m$.

Remark: This result is called, for historical reasons, the *Invariance of Domain Theorem*. For the proof, use Exercises 12 and 13.

5. Degrees of Maps

1. Fix an integer k and define a function $f : S^1 \to S^1$ by $f(e^{i\theta}) = e^{2ki\theta}$ if $0 \leq \theta < \pi$ and $f(e^{i\theta}) = e^{-2ki\theta}$ if $\pi \leq \theta < 2\pi$. Show that f is continuous and homotopic to a constant map. For each $e^{i\theta} \in S^1$, how many points are there in $f^{-1}(\{e^{i\theta}\})$?

2. Suppose $f : S^1 \to S^1$ is a C^∞ mapping such that (for convenience) $f((1,0)) = (1,0)$. Suppose also that $\dfrac{d}{d\theta} \arg f(e^{i\theta})$ vanishes at only finitely many points of S^1. Suppose finally that f is homotopic to a constant mapping. Prove that, for any $p \in S^1$ that is not the image under f of one of the (finitely many) points at which $\dfrac{d}{d\theta} \arg f(e^{i\theta}) = 0$, the number of points in $f^{-1}(\{p\})$ is finite. Prove also that for such p the number of points in $f^{-1}(\{p\})$ at which $\dfrac{d}{d\theta} \arg f(e^{i\theta}) > 0$ equals the number of points of $f^{-1}(\{p\})$ at which $\dfrac{d}{d\theta} \arg f(e^{i\theta}) < 0$. *Hint:* Think of f as a map from $[0,1]$ to S^1 with $f(0) = f(1)$. Then consider the "lift" of f to a map $h : [0,1] \to \mathbb{R}$, where $e^{ih(t)} = f(t)$, as in the discussion of covering spaces in Section III.5. Then $h(0) = h(1) = 0$ because h is homotopic to a constant. Now look at the graph of h and count the number of points where the graph hits a fixed horizontal line (or a fixed finite collection of horizontal lines that are 2π apart).



3. Construct a map f of S^1 to S^1 such that $f^{-1}(\{q\})$ is an infinite set for every $q \in S^1$. *Hint:* Recall the Peano curve $h : [0,1] \to [0,1] \times [0,1]$ described in Exercise II.6.11. Consider maps of the form $f(e^{i\theta}) = e^{2\pi i g}$, where g is the x-component of $h(\theta/2\pi)$.

4. Establish the following for a C^∞ map from S^1 to S^1:

(a) If $f^{-1}(\{p\})$ is infinite for some $p \in S^1$, then there is a point of $f^{-1}(\{p\})$ at which $\dfrac{d}{d\theta} \arg f(e^{i\theta})$ vanishes. *Hint:* Look at a limit point of a sequence of distinct points of $f^{-1}(\{p\})$.

(b) If A is the set of points in S^1 at which $\dfrac{d}{d\theta} \arg f(e^{i\theta})$ vanishes, then $f(A)$ is a closed set with empty interior. *Hint:* Cover A with a finite number of open arcs of total length at most 2π such that $\left| \dfrac{d}{d\theta} \arg f(e^{i\theta}) \right| < \varepsilon/2\pi$ at every point of each of the arcs. Then obtain a cover of $f(A)$ by a finite number of arcs of total length less than ε.

5. Show that $f^{-1}(\{q\})$ is finite if q is a noncritical point of a C^∞ map $f : S^n \to S^n$. *Hint:* Show that $f^{-1}(\{q\})$ has no limit points by applying the Inverse Function Theorem to the radial extension $\widehat{f}$ of f.

6. Show that if there is a tangent vector field $V : S^n \to \mathbb{R}^{n+1}$ on S^n, then the antipodal map is homotopic to the identity map of S^n to S^n. *Hint:* In the case $|V| = 1$ on S^n, consider

$$H(p,t) = (\cos(\pi t))p + (\sin(\pi t))V(p), \qquad p \in S^n, 0 \leq t \leq 1.$$

7. Prove that the set of critical points of a C^∞ map $f : S^n \to S^n$ is closed.

8. Suppose that f is a C^∞ map from S^n to S^n and that $\widehat{f}$ is the extension of f, defined in this section. Prove that the Jacobian determinant J of $\widehat{f}$ satisfies $J(\lambda v) = J(v)$ for all positive $\lambda \in \mathbb{R}$ and all $v \in \mathbb{R}^{n+1} \setminus \{(0, \ldots, 0)\}$. Deduce that the set of critical points of f on S^n has empty interior in S^n if and only if the set of critical points of $\widehat{f}$ (i.e., the image under $\widehat{f}$ of the zero set of J) has empty interior in $\mathbb{R}^{n+1}$.

9. Prove that if $f, g : S^n \to S^n$ are C^∞ maps that are homotopic (as continuous maps), then they are C^∞-homotopic, i.e., there is a C^∞ map $H : S^n \times [0,1] \to S^n$ such that $H(x,0) = f(x)$ and $H(x,1) = g(x)$ for all $x \in S^n$. *Hint:* Using Exercise 2.2, first construct an auxiliary C^∞ function $\rho_\varepsilon : [0,1] \to [0,1]$, for $\varepsilon > 0$ small, such that $\rho_\varepsilon(t) = 0$ if $\varepsilon \le t \le 1-\varepsilon$ and $\rho_\varepsilon(t) = 1$ if $0 \le t \le \varepsilon/2$ or if $1-\varepsilon/2 \le t \le 1$. Then let F be a continuous homotopy of f and g, approximate F by a function $G(x,t)$ that is a polynomial in $x_1, \ldots, x_n$ and t, and set $H_0(x,t) = \rho_\varepsilon(t)f(x) + [1 - \rho_\varepsilon(t)]G(x,t)$, $H(x,t) = H_0(x,t)/|H_0(x,t)|$, for $x \in S^n$ and $0 \le t \le 1/2$, with a similar formula for $1/2 \le t \le 1$.

10. Let $U, V \subseteq \mathbb{R}^n$ be connected open sets and let $f : U \to V$ be a homeomorphism. For each p in U, choose an $\varepsilon > 0$ such that $B(p; 2\varepsilon) \subseteq U$. Consider the open set $f(B(p; 2\varepsilon)) \subseteq V$ and choose $\delta > 0$ such that $B(f(p), 2\delta) \subseteq f(B(p; 2\varepsilon))$. Define a map $\sigma : S^n \to B(p; 2\varepsilon) \setminus \{p\}$ by $\sigma(x) = f^{-1}(f(p) + \delta x)$.

(a) Show that $\sigma : S^n \to B(p; 2\varepsilon) \setminus \{p\}$ is not homotopic to a constant map of S^n into $B(p; 2\varepsilon) \setminus \{p\}$. Deduce that the degree of the map $x \mapsto (\sigma(x) - p)/\|\sigma(x) - p\|$ is not zero.

(b) Show that this degree is independent of the choices of δ and ε.

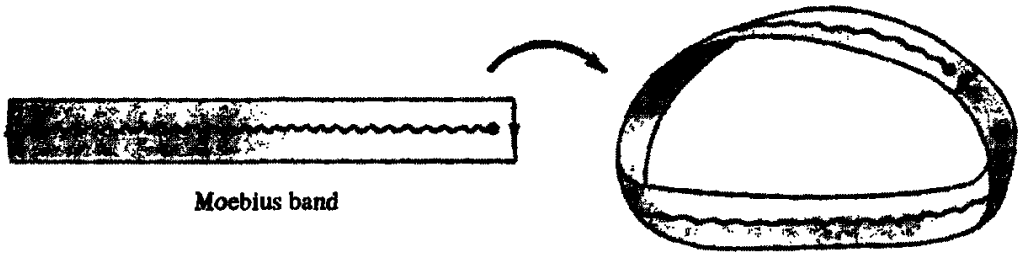
Remark: The map f is *orientation-preserving* at p if this degree is positive and *orientation-reversing* at p if this degree is negative.

(c) Show that either f is orientation-preserving at every point of U or f is orientation-reversing at every point of U.

(d) Show that the composition of two orientation-preserving homeomorphisms is orientation-preserving and that the inverse of an orientation-preserving homeomorphism is orientation-preserving.

(e) Show that if f is C^∞ with a C^∞ inverse, then f is orientation-preserving at p if and only if the Jacobian determinant of f at p is positive.

11. Let X be the quotient space obtained from the semiopen square $\{(x, y) : 0 \leq x \leq 1, 0 < y < 1\}$ by identifying $(0, y)$ with $(1, 1 - y)$, $0 < y < 1$. Prove that X is a differentiable manifold. Prove that X is not orientable.



Remark: The manifold X is known as a *Moebius band*.

12. Prove that S^n is a differentiable manifold and that it is orientable. *Hint:* To coordinatize open subsets of S^n, consider maps from subsets of S^n to $\mathbb{R}^n$ of the form $(x_1, \ldots, x_{n+1}) \mapsto (x_1, \ldots, x_{k-1}, x_{k+1}, \ldots, x_{n+1})$.

13. Suppose that M is a differentiable manifold, $\{U_\alpha\}_{\alpha\in A}$ is an open cover of M, and $\psi_\alpha : U_\alpha \to \mathbb{R}^n$ are homeomorphisms such that $\psi_\alpha \circ \psi_\beta^{-1}$ is a C^∞ map where defined (on $\psi_\alpha(U_\alpha \cap U_\beta)$). A *vector field on M* is a collection of vector fields $V_\alpha : \psi_\alpha(U_\alpha) \to \mathbb{R}^n$ on the open sets $\psi_\alpha(U_\alpha)$ with the property that for every $\alpha, \beta \in A$ and every $p \in \psi_\alpha(U_\alpha \cap U_\beta)$, the image of $V_\alpha(p)$ under the Jacobian linear mapping of $\psi_\beta \circ \psi_\alpha^{-1}$ is $V_\beta(\psi_\beta(\psi_\alpha^{-1}(p)))$. Show that a tangent vector field on S^n in the sense of the text is associated to a tangent vector field on S^n considered as a differentiable manifold as in Exercise 12 by the following association. If $V : S^n \to \mathbb{R}^{n+1}$ is a tangent vector field (in the sense of the text) and if $\psi : U \to \mathbb{R}^{n+1}$, $U \subseteq S^n$, is a coordinatizing homeomorphism, then set $V_\psi(p)$, $p \in \psi(U)$, equal to the tangent vector at $t = 0$ to the curve $\gamma(t)$ defined by $\gamma(t) = \psi(\gamma_1(t))$, where $\gamma_1(t) = (p + tV(p))/\|p + tV(p)\|$.

14. Show that the Euler characteristic of S^n is zero if n is odd and 2 if n is even. *Hint:* For n odd, there is a tangent vector field on S^n without zeros. For n even, consider the tangent vector field V defined by

$$V(x_1, \ldots, x_{n+1}) = (-x_1 x_{n+1}, \ldots, -x_n x_{n+1}, 1 - x_{n+1}^2), \qquad x \in S^n.$$

Show that it is a tangent vector field and find the indices of its zeros.